

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
                dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7874

Nombre de lignes df clean	7874
Nombre de features	68
% de NaN	0.0
Taille dataset entraînement	5511
Taille dataset test	2362
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms, r_NH3_wl_24h_D1_energy_rel, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_NH3_wl_24h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy, r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D2_energy, r_umidity_wl_24h_D3_energy, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms, r_umidity_wl_24h_D1_energy_rel, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

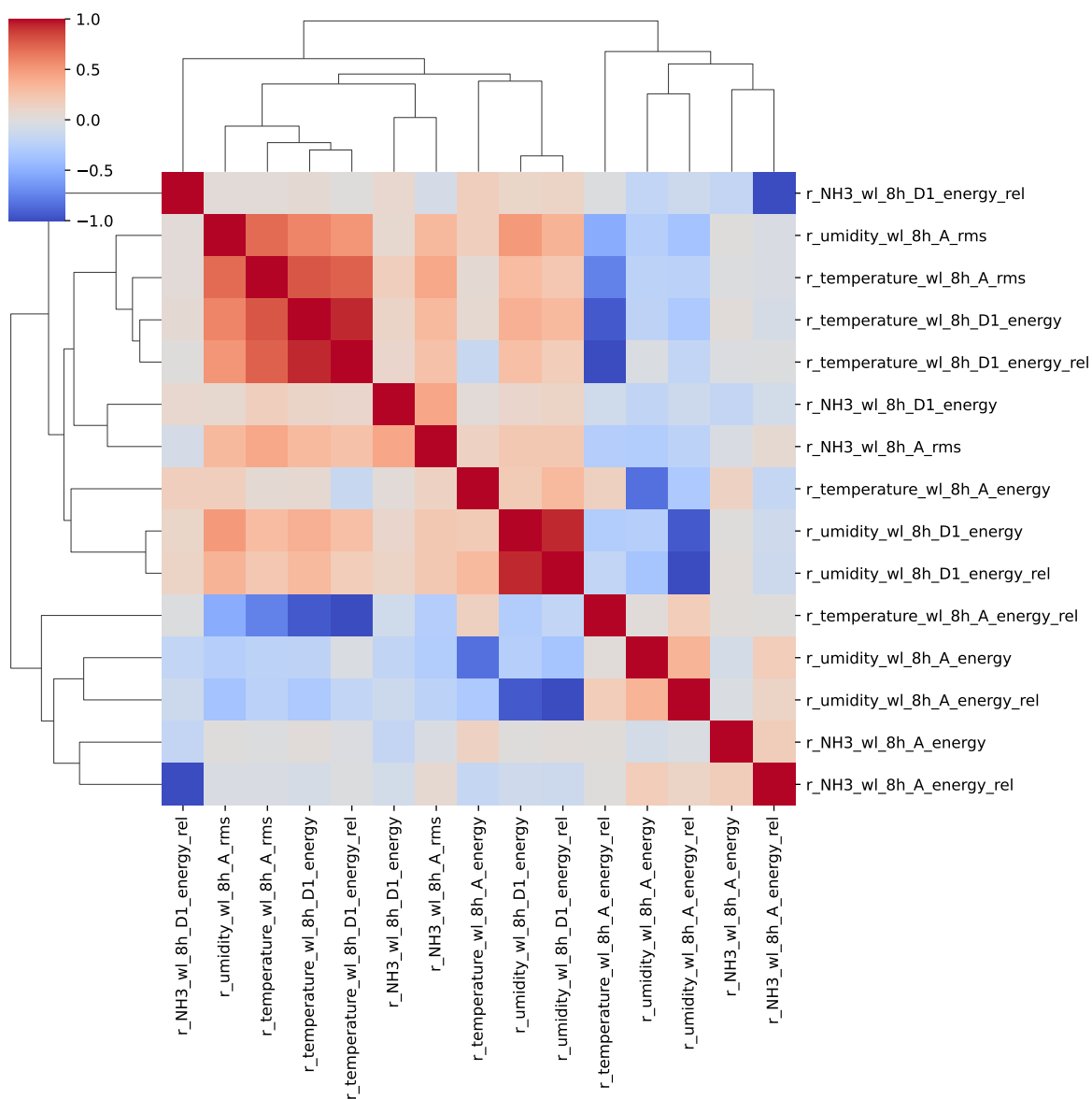
The following section has been done with a threshold choosen of : 0.8

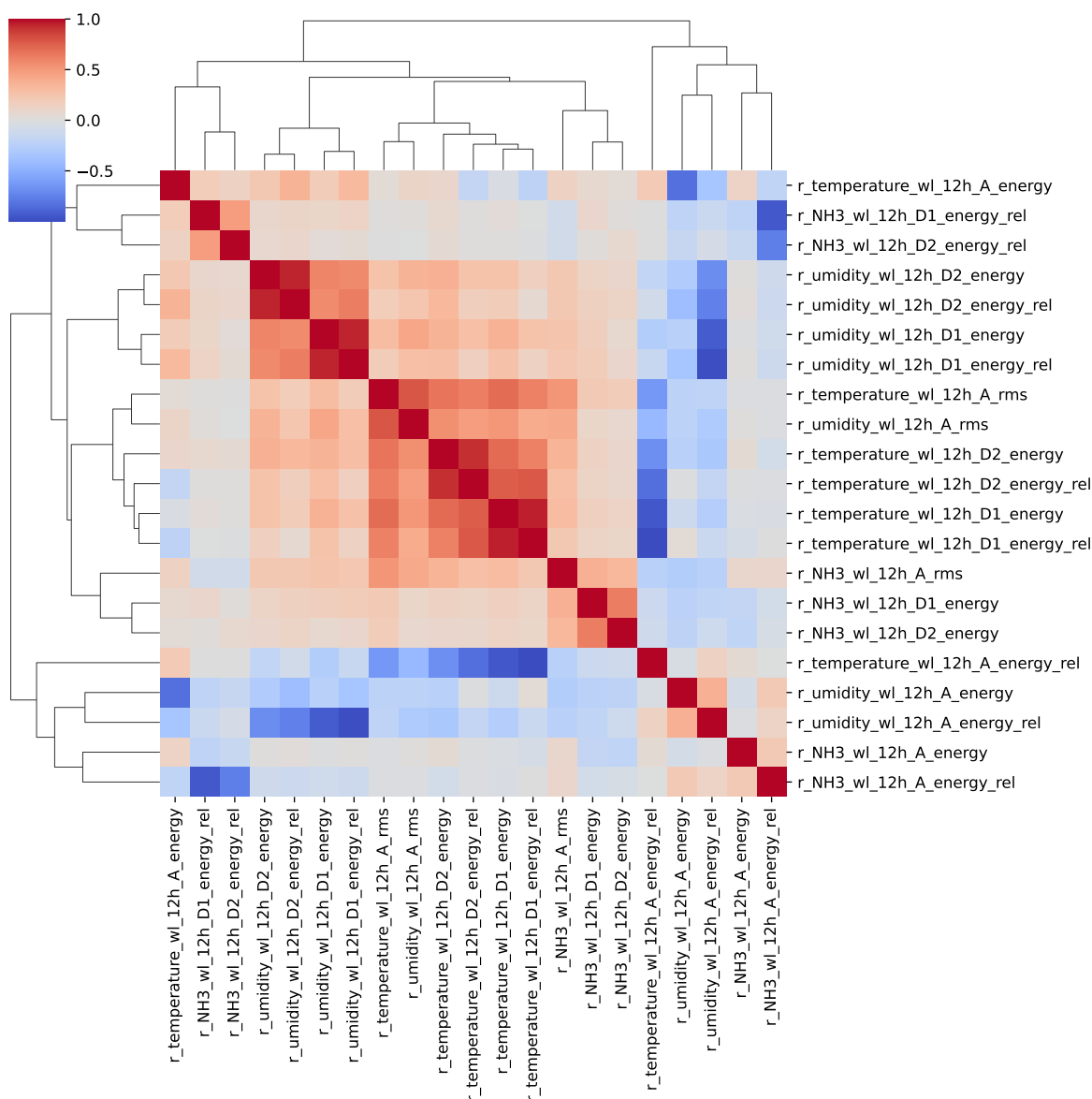
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_wl_8h_D1_energy ; **cluster 7:** r_NH3_wl_8h_A_energy ; **cluster 8:** r_NH3_wl_8h_A_rms ; **cluster 9:** r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_A_energy_rel ; **cluster 10:** r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D1_energy_rel ; **cluster 11:** r_temperature_wl_8h_A_energy, r_umidity_wl_8h_A_energy ; **cluster 12:** r_temperature_wl_8h_A_rms ; **cluster 13:** r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy ; **cluster 14:** r_umidity_wl_8h_A_rms ; **cluster 15:** r_NH3_wl_12h_D1_energy ; **cluster 16:** r_NH3_wl_12h_D2_energy ; **cluster 17:** r_NH3_wl_12h_A_energy ; **cluster 18:** r_NH3_wl_12h_A_rms ; **cluster 19:** r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_A_energy_rel ; **cluster 20:** r_NH3_wl_12h_D2_energy_rel ; **cluster 21:** r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D1_energy_rel ; **cluster 22:** r_temperature_wl_12h_A_energy, r_umidity_wl_12h_A_energy ; **cluster 23:** r_temperature_wl_12h_A_rms ; **cluster 24:** r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D1_energy_rel ; **cluster 25:** r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D2_energy ; **cluster 26:** r_umidity_wl_12h_A_rms ; **cluster 27:** r_NH3_wl_24h_D1_energy ; **cluster 28:** r_NH3_wl_24h_D2_energy ; **cluster 29:** r_NH3_wl_24h_D3_energy ; **cluster 30:** r_NH3_wl_24h_A_energy ; **cluster 31:** r_NH3_wl_24h_A_rms ; **cluster 32:** r_NH3_wl_24h_A_energy_rel, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_D1_energy_rel ; **cluster 33:** r_NH3_wl_24h_D3_energy_rel ; **cluster 34:** r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_A_energy_rel, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D2_energy ; **cluster 35:** r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_D3_energy ; **cluster 36:** r_umidity_wl_24h_A_energy, r_temperature_wl_24h_A_energy ; **cluster 37:** r_umidity_wl_24h_A_rms, r_temperature_wl_24h_A_rms ; **cluster 38:** r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_A_energy_rel, r_umidity_wl_24h_D1_energy_rel ; **cluster 39:** r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D2_energy ; **cluster 40:** r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_D3_energy ;

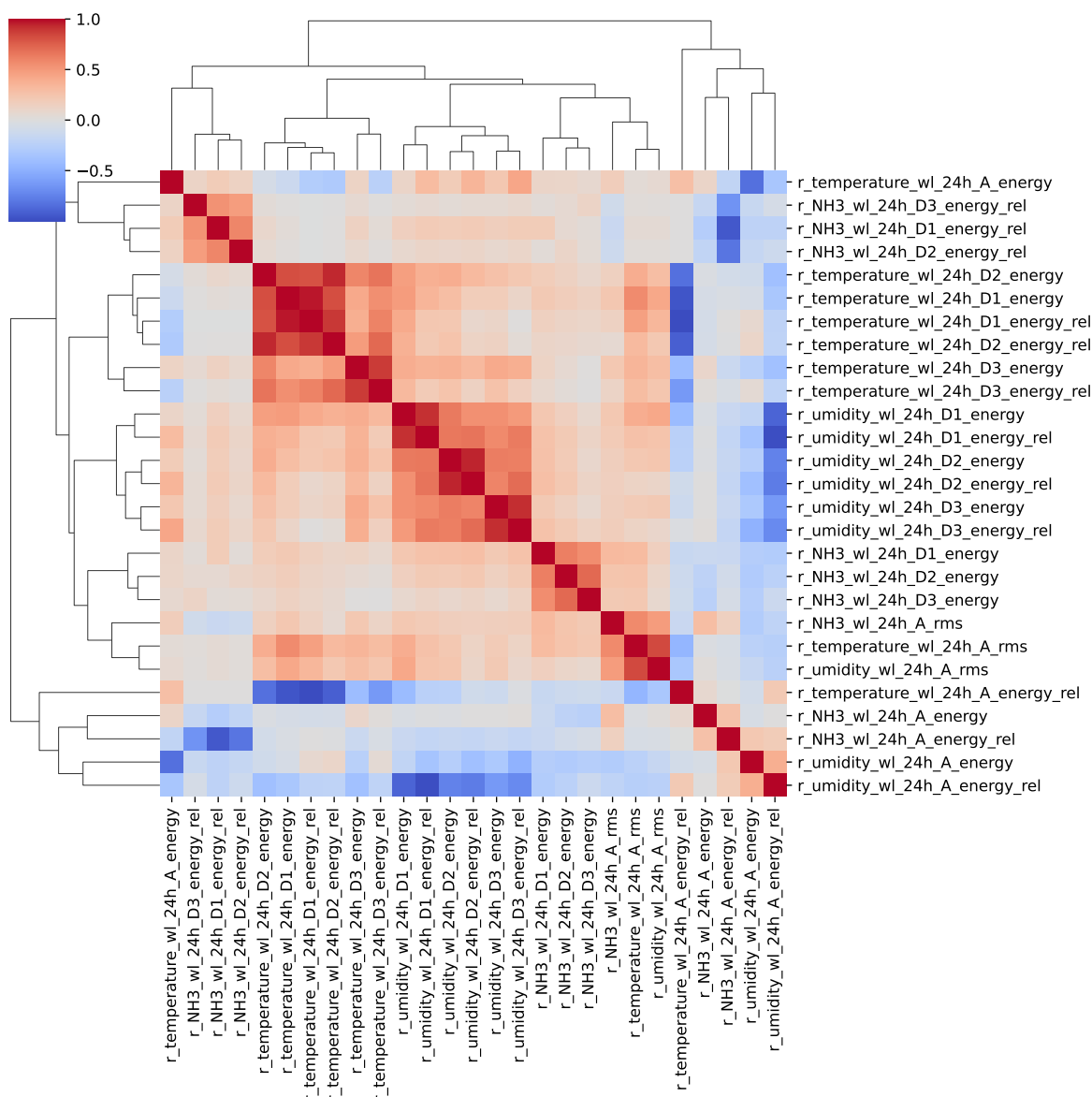
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_A_rms, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms, r_NH3_wl_24h_A_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D3_energy_rel







3 Model(s) analysis

3.1 Metadata

Features X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel,

r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_A_rms, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms, r_NH3_wl_24h_A_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D3_energy_rel

Len(Features_X): 40

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

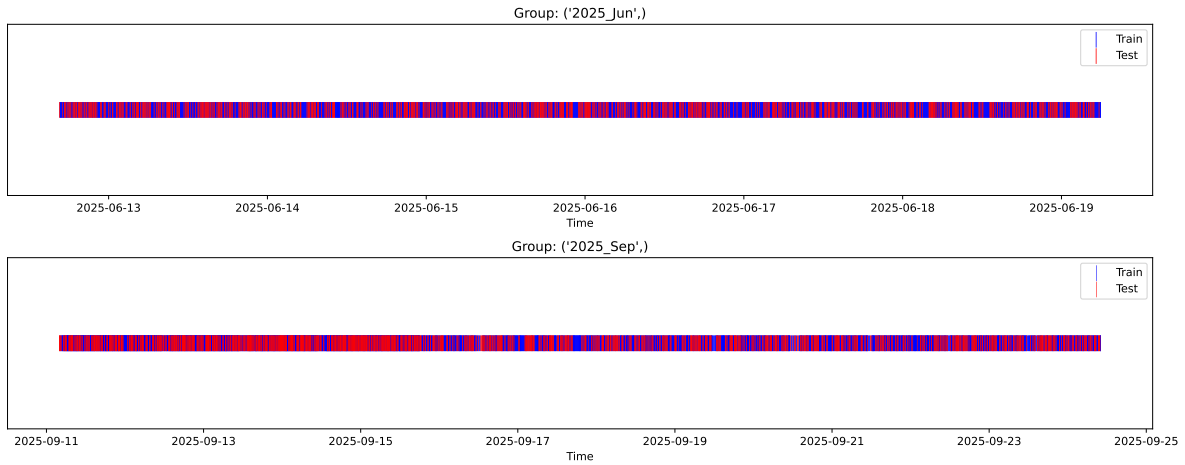
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5511 samples Test: 2363 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3857 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3857 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3857 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3857 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3857 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.804, min=0.959, max=13.845
 y_{test} : mean=2.342, std=0.779, min=1.062, max=7.779

Fold 0 — y_{test} : mean=2.35, std=0.87, y_{train} : mean=2.34, std=0.77

Fold 1 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.34, std=0.81

Fold 2 — y_{test} : mean=2.31, std=0.74, y_{train} : mean=2.36, std=0.83

Fold 3 — y_test: mean=2.32, std=0.79, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.31, std=0.76, y_train: mean=2.36, std=0.82

train scores CV: [0.57761428 0.55961853 0.56856363 0.56270943 0.5559948]

test scores CV: [0.54277181 0.52946189 0.5528727 0.5670153 0.58703753]

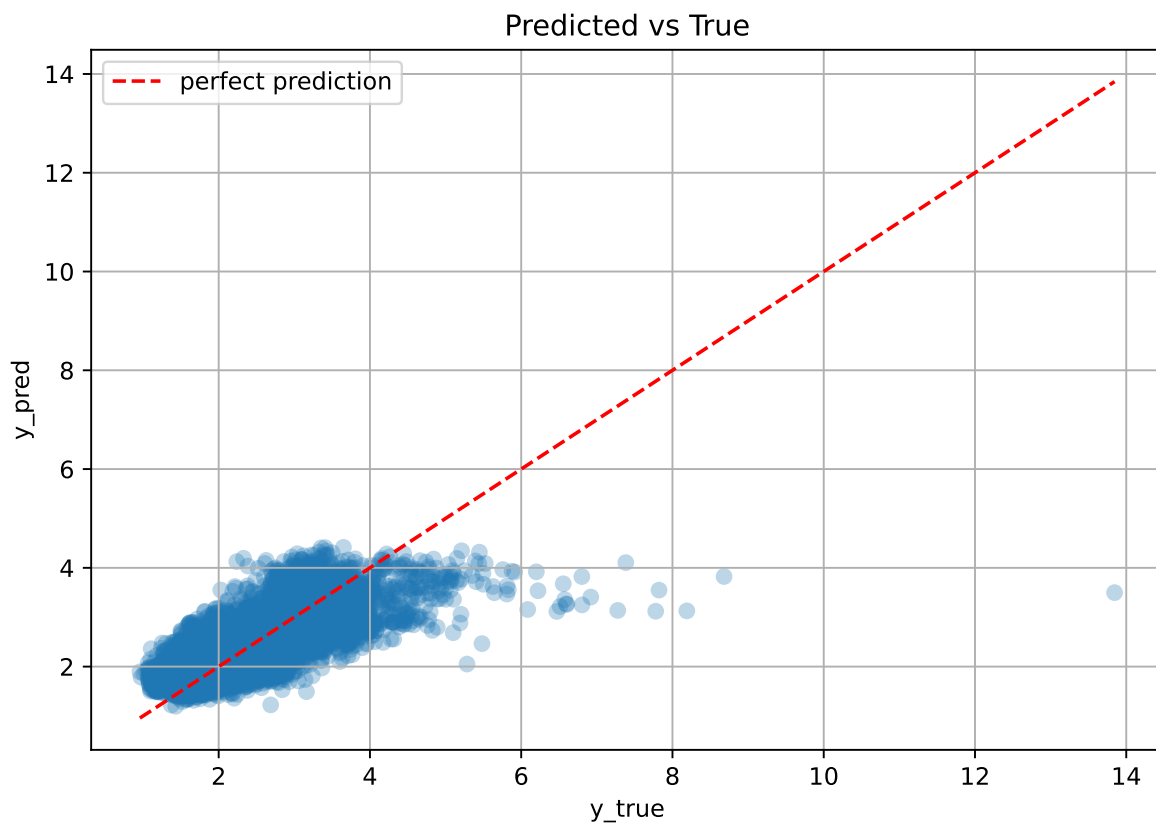
rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5649	0.5558	0.0199	0.5566	0.5603	0.0045	1.0081	1.0163

Coefficients:

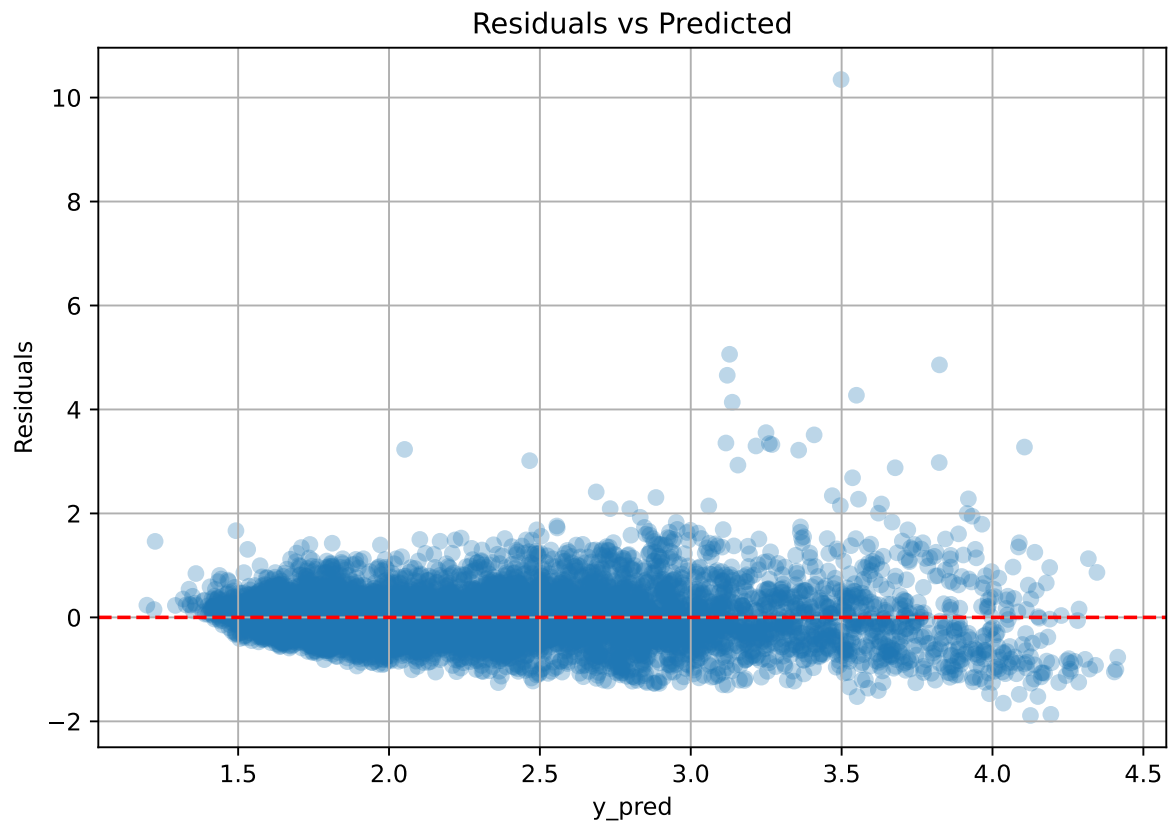
coef_r_CO2 = -0.0472, coef_r_NH3 = 0.2742, coef_r_NH3_wl_12h_A_energy = 0.2019, coef_r_NH3_wl_12h_A_rms = 0.0370, coef_r_NH3_wl_12h_D1_energy = -0.0213, coef_r_NH3_wl_12h_D1_energy_rel = -0.0256, coef_r_NH3_wl_12h_D2_energy = 0.0107, coef_r_NH3_wl_12h_D2_energy_rel = -0.0381, coef_r_NH3_wl_24h_A_energy = -0.3265, coef_r_NH3_wl_24h_A_energy_rel = 0.0423, coef_r_NH3_wl_24h_A_rms = -0.0294, coef_r_NH3_wl_24h_D1_energy = 0.0253, coef_r_NH3_wl_24h_D2_energy = 0.0906, coef_r_NH3_wl_24h_D3_energy = -0.0763, coef_r_NH3_wl_24h_D3_energy_rel = 0.0154, coef_r_NH3_wl_8h_A_energy = -0.2050, coef_r_NH3_wl_8h_A_rms = 0.0121, coef_r_NH3_wl_8h_D1_energy = 0.0025, coef_r_NH3_wl_8h_D1_energy_rel = 0.0355, coef_r_THI = 0.4719, coef_r_temperature = 0.1977, coef_r_temperature_wl_12h_A_energy = -0.2542, coef_r_temperature_wl_12h_A_rms = 0.1715, coef_r_temperature_wl_12h_D2_energy = 0.0245, coef_r_temperature_wl_24h_D1_energy_rel = -0.1064, coef_r_temperature_wl_24h_D3_energy_rel = -0.0153, coef_r_temperature_wl_8h_A_energy = -0.4516, coef_r_temperature_wl_8h_A_rms = -0.0067, coef_r_temperature_wl_8h_D1_energy = -0.0195, coef_r_umidity = -0.6400, coef_r_umidity_wl_12h_A_rms = 0.0120, coef_r_umidity_wl_12h_D1_energy = 0.0180, coef_r_umidity_wl_12h_D2_energy_rel = 0.0227, coef_r_umidity_wl_24h_A_energy = 0.1676, coef_r_umidity_wl_24h_A_rms = -0.1043, coef_r_umidity_wl_24h_D1_energy = -0.0337, coef_r_umidity_wl_24h_D2_energy_rel = -0.0228, coef_r_umid-

ity_wl_24h_D3_energy_rel = 0.0430, coef_r_umidity_wl_8h_A_rms = 0.0541,
coef_r_umidity_wl_8h_D1_energy_rel = -0.0301, intercept = 2.3436,

3.4.2 Scatter



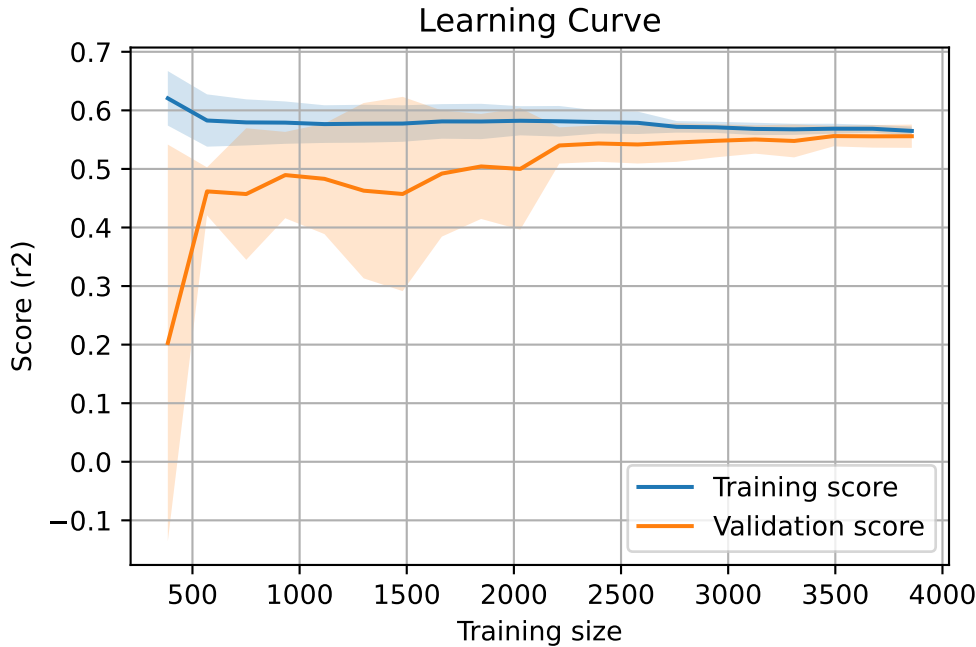
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7874, `Len(training)` : 5512,

`Len(training_real)` : 3859, `Len(test_of_training)` : 1653, `Len(test)` : 2362



3.5 Model : LGBM

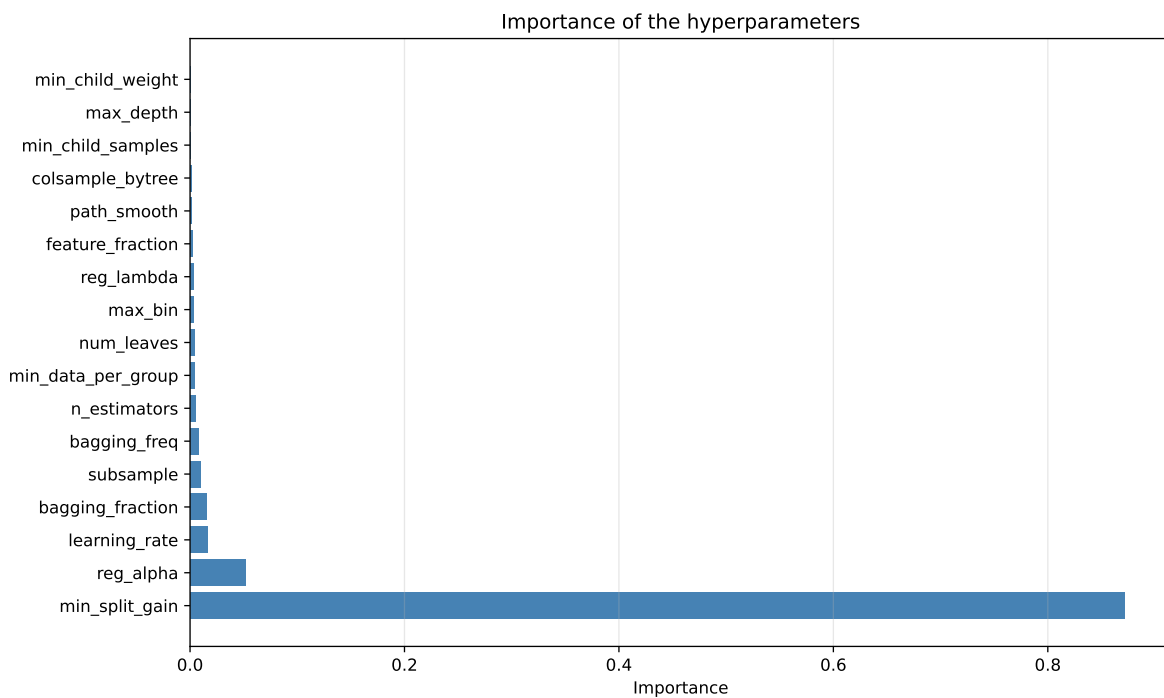
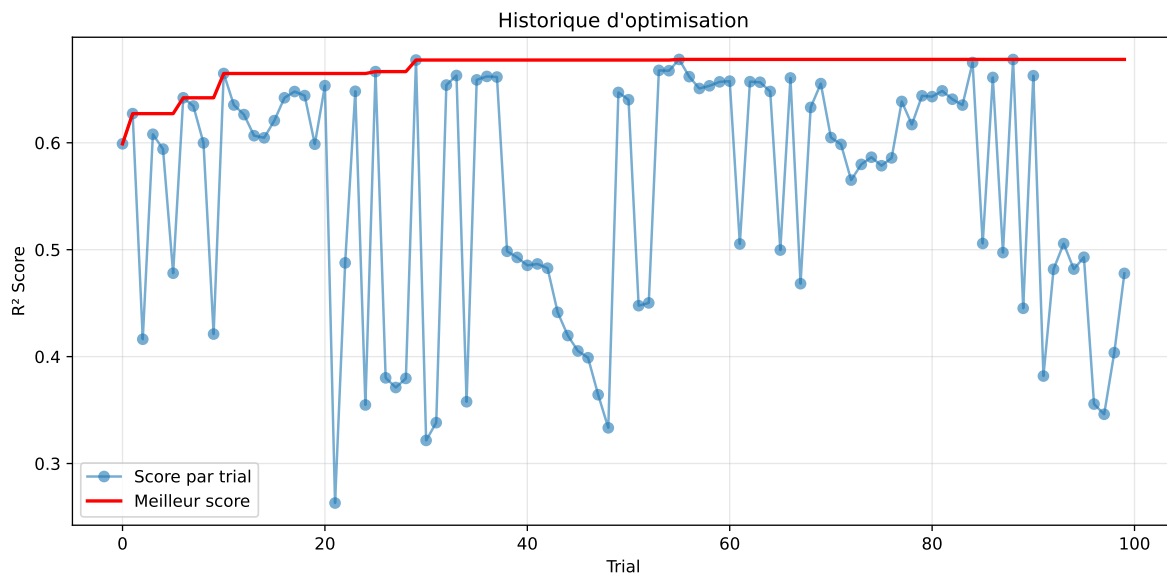
3.5.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.804, min=0.959, max=13.845
 y_{test} : mean=2.342, std=0.779, min=1.062, max=7.779

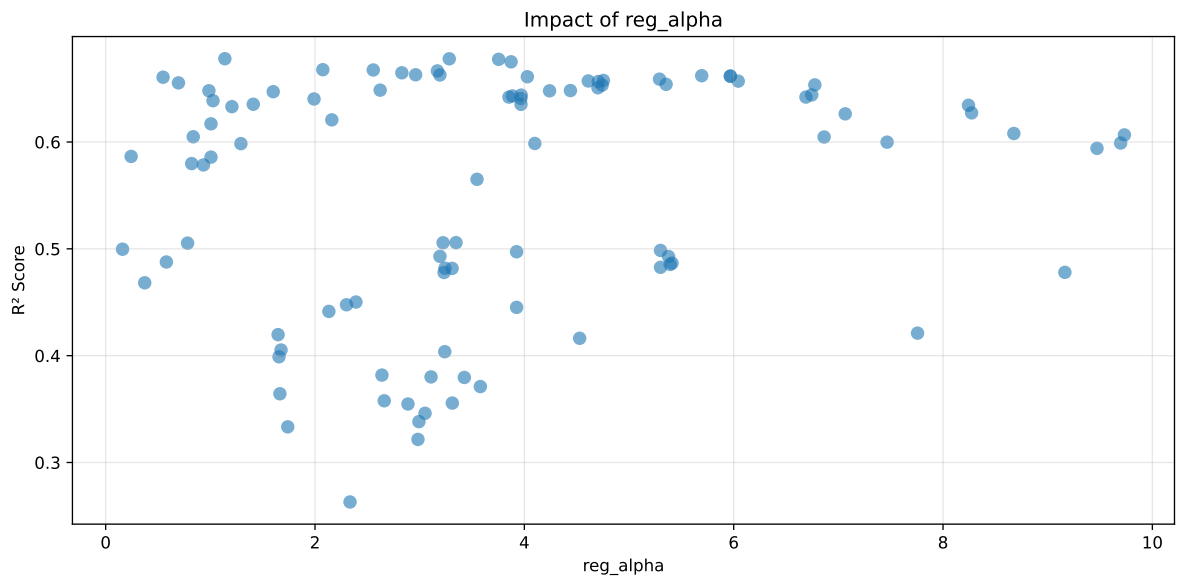
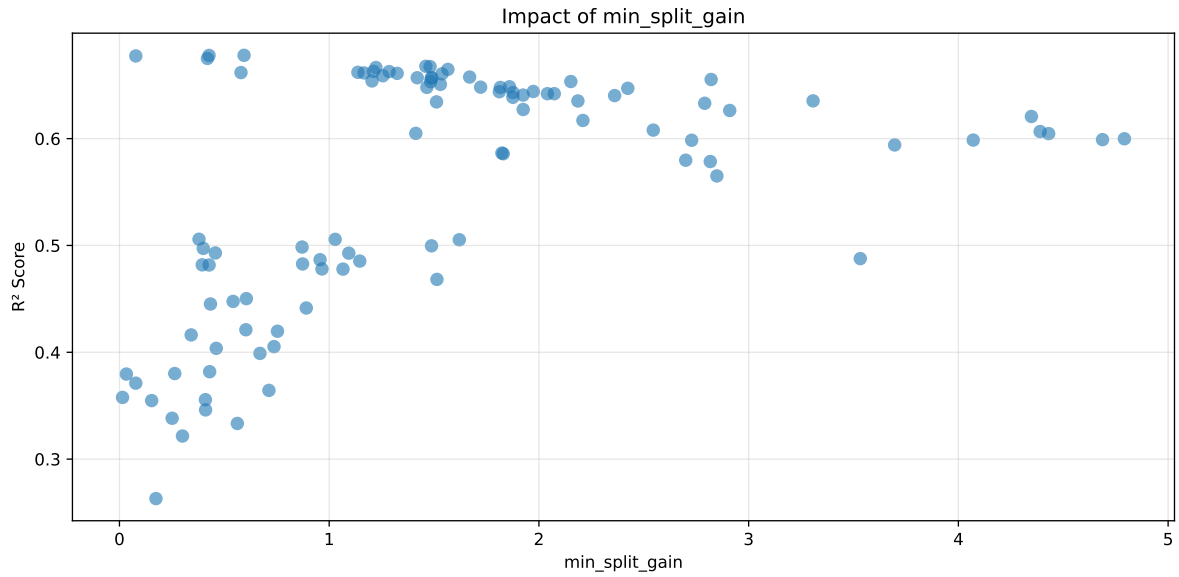
===== TOP Importance FEATURES =====

feature	importance
r_CO2	549
r_umidity	403
r_NH3	254
r_NH3_wl_24h_A_en- ergy	200
r_temperature_wl_24h_D1_energy_rel	192
r_temperature	189
r_temperature_wl_12h_A_rms	177
r_NH3_wl_24h_D3_energy_rel	168
r_umidity_wl_24h_A_en- ergy	132
r_umidity_wl_12h_A_rms	127
r_THI	124
r_NH3_wl_24h_D3_energy	119
r_NH3_wl_24h_D1_energy	118
r_NH3_wl_24h_A_rms	117
r_temperature_wl_12h_A_en- ergy	113
r_temperature_wl_8h_A_rms	105
r_umidity_wl_24h_D1_energy	99
r_NH3_wl_24h_D2_energy	97
r_temperature_wl_24h_D3_energy_rel	96
r_NH3_wl_12h_A_en- ergy	84
r_umidity_wl_24h_D2_energy_rel	83
r_NH3_wl_8h_A_rms	81
r_tempera- ture_wl_8h_A_energy	79
r_NH3_wl_8h_A_energy	74
r_umidity_wl_12h_D2_en- ergy_rel	69
r_umidity_wl_12h_D1_energy	55
r_umidity_wl_24h_A_rms	53
r_umid- ity_wl_8h_D1_energy_rel	51
r_umidity_wl_8h_A_rms	48
r_NH3_wl_12h_D2_energy	48
r_NH3_wl_8h_D1_energy	39
r_umidity_wl_24h_D3_energy_rel	37
r_NH3_wl_24h_A_en- ergy_rel	37
r_NH3_wl_12h_D2_energy_rel	32
r_NH3_wl_12h_A_rms	25
r_tempera-	

ture_wl_12h_D2_energy 22 r_NH3_wl_12h_D1_energy 21 r_NH3_wl_8h_D1_energy_rel
20 r_temperature_wl_8h_D1_energy 19 r_NH3_wl_12h_D1_energy_rel 12

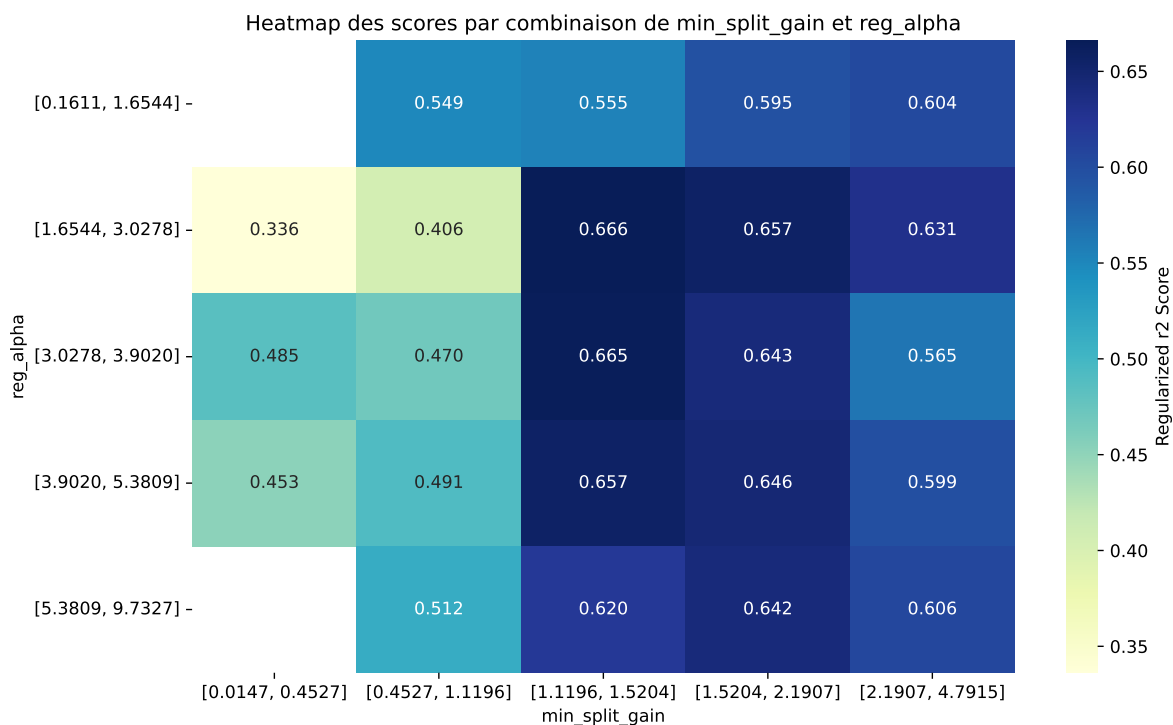


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7113	0.678	0.6732	0.0226	0.678	0.6893	0.0113	1.0167	1.0491
2	0.7111	0.6779	0.672	0.0203	0.6779	0.6814	0.0035	1.0052	1.049
3	0.7107	0.6774	0.6723	0.0196	0.6774	0.6842	0.0068	1.01	1.0492
4	0.7078	0.6751	0.6702	0.0187	0.6751	0.6807	0.0056	1.0083	1.0484
5	0.701	0.6677	0.6643	0.0216	0.6677	0.6843	0.0166	1.0249	1.0498
6	0.6984	0.6674	0.6625	0.0229	0.6674	0.6884	0.021	1.0315	1.0465
7	0.6946	0.6666	0.6588	0.021	0.6666	0.6811	0.0146	1.0219	1.0421
8	0.6947	0.6648	0.6595	0.0215	0.6648	0.6809	0.0161	1.0242	1.045
9	0.6926	0.6629	0.6568	0.0204	0.6629	0.6816	0.0186	1.0281	1.0447
10	0.6876	0.6627	0.6544	0.02	0.6627	0.6732	0.0105	1.0158	1.0376
---	---	---	---	---	---	---	---	---	---
100	0.8246	0.731	0.7322	0.0262	0.2631	0.7488	0.0178	1.0243	1.128

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.013506723224766003, 'reg__num_leaves': 36, 'reg__subsample': 0.7854635424892356, 'reg__colsample_bytree': 0.7775904730584067, 'reg__min_child_samples': 48, 'reg__reg_alpha': 1.1387501287762507, 'reg__reg_lambda': 34.738903046763966, 'reg__min_split_gain': 0.594148036458284, 'reg__path_smooth': 0.986176425322532, 'reg__min_child_weight':

0.06774421360541688, 'reg__feature_fraction': 0.7343484162671667, 'reg__bagging_fraction': 0.5972337316634915, 'reg__bagging_freq': 6, 'reg__max_bin': 234, 'reg__min_data_per_group': 159}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.01650066229138209, 'reg__num_leaves': 50, 'reg__subsample': 0.709193111942329, 'reg__colsample_bytree': 0.715620840830008, 'reg__min_child_samples': 44, 'reg__reg_alpha': 3.2825763807552026, 'reg__reg_lambda': 31.103790765113434, 'reg__min_split_gain': 0.4273533359509748, 'reg__path_smooth': 1.9610089587357273, 'reg__min_child_weight': 0.015980206220146256, 'reg__feature_fraction': 0.885136463203861, 'reg__bagging_fraction': 0.6714400125964628, 'reg__bagging_freq': 5, 'reg__max_bin': 244, 'reg__min_data_per_group': 173}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.016303471815355228, 'reg__num_leaves': 45, 'reg__subsample': 0.7464337065689106, 'reg__colsample_bytree': 0.891276645500813, 'reg__min_child_samples': 43, 'reg__reg_alpha': 3.7544951449628092, 'reg__reg_lambda': 28.207615024979653, 'reg__min_split_gain': 0.07802139164450494, 'reg__path_smooth': 0.10857145104958214, 'reg__min_child_weight': 0.013663027471771451, 'reg__feature_fraction': 0.7122758639775415, 'reg__bagging_fraction': 0.6447880013131011, 'reg__bagging_freq': 3, 'reg__max_bin': 241, 'reg__min_data_per_group': 169}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.01665064582715127, 'reg__num_leaves': 50, 'reg__subsample': 0.7363887987343871, 'reg__colsample_bytree': 0.8466993925420073, 'reg__min_child_samples': 44, 'reg__reg_alpha': 3.8733672619596633, 'reg__reg_lambda': 36.69733058290834, 'reg__min_split_gain': 0.41972109619915443, 'reg__path_smooth': 1.2522276929848712, 'reg__min_child_weight': 0.04670778596346991, 'reg__feature_fraction': 0.87589422849114, 'reg__bagging_fraction': 0.6479121406290581, 'reg__bagging_freq': 3, 'reg__max_bin': 245, 'reg__min_data_per_group': 190}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.02121488818985323, 'reg__num_leaves': 36, 'reg__subsample': 0.7944904256995552, 'reg__colsample_bytree': 0.7876998600334506, 'reg__min_child_samples': 36, 'reg__reg_alpha': 2.074676657327705, 'reg__reg_lambda': 34.567265779429235, 'reg__min_split_gain': 1.4608732128985522, 'reg__path_smooth': 0.9880985867960136, 'reg__min_child_weight': 0.07811982974470769, 'reg__feature_fraction': 0.7326533632144956, 'reg__bagging_fraction': 0.6917300722603865, 'reg__bagging_freq': 5, 'reg__max_bin': 207, 'reg__min_data_per_group': 164}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.013193210406893849, 'reg__num_leaves': 38, 'reg__subsample': 0.7952200765087793, 'reg__colsample_bytree': 0.7712017498390122, 'reg__min_child_samples': 32, 'reg__reg_alpha': 2.5564358341926474, 'reg__reg_lambda': 26.339690083892055, 'reg__min_split_gain': 1.4811038435181387, 'reg__path_smooth': 1.0019846606187404, 'reg__min_child_weight':

0.0011685186153087247, 'reg__feature_fraction': 0.816064393047327, 'reg__bagging_fraction': 0.6014662917984891, 'reg__bagging_freq': 5, 'reg__max_bin': 144, 'reg__min_data_per_group': 160}

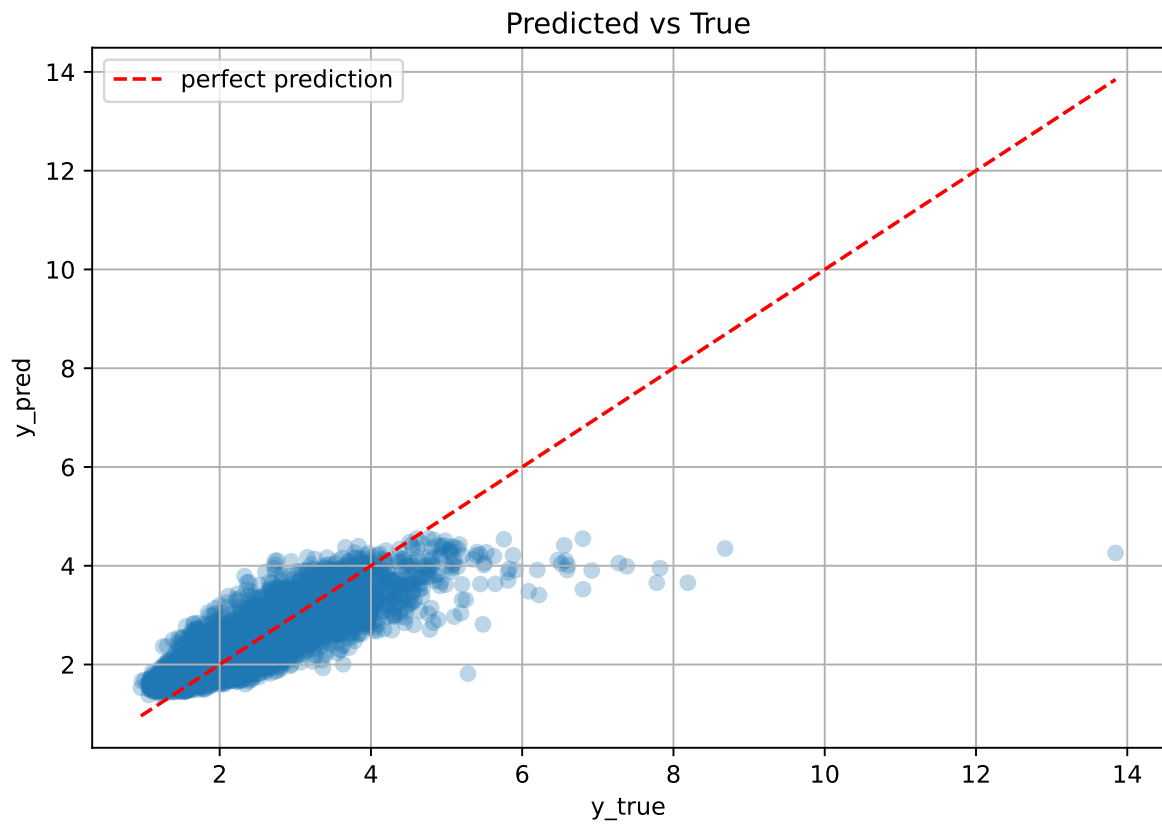
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.03536026876714366, 'reg__num_leaves': 51, 'reg__subsample': 0.5428562145974403, 'reg__colsample_bytree': 0.7558938227196149, 'reg__min_child_samples': 46, 'reg__reg_alpha': 3.1690774169719402, 'reg__reg_lambda': 30.434212444067647, 'reg__min_split_gain': 1.2228591090774419, 'reg__path_smooth': 0.32868288826587333, 'reg__min_child_weight': 0.03301491280636847, 'reg__feature_fraction': 0.7193480166889289, 'reg__bagging_fraction': 0.6703205850930312, 'reg__bagging_freq': 3, 'reg__max_bin': 254, 'reg__min_data_per_group': 188}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.024343465341787794, 'reg__num_leaves': 38, 'reg__subsample': 0.7189620494906236, 'reg__colsample_bytree': 0.8350088814698102, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.829833104935134, 'reg__reg_lambda': 32.75850330849816, 'reg__min_split_gain': 1.565933542251487, 'reg__path_smooth': 0.09675141135376375, 'reg__min_child_weight': 0.031020007928956362, 'reg__feature_fraction': 0.7492924091950717, 'reg__bagging_fraction': 0.6430285098779241, 'reg__bagging_freq': 5, 'reg__max_bin': 226, 'reg__min_data_per_group': 180}

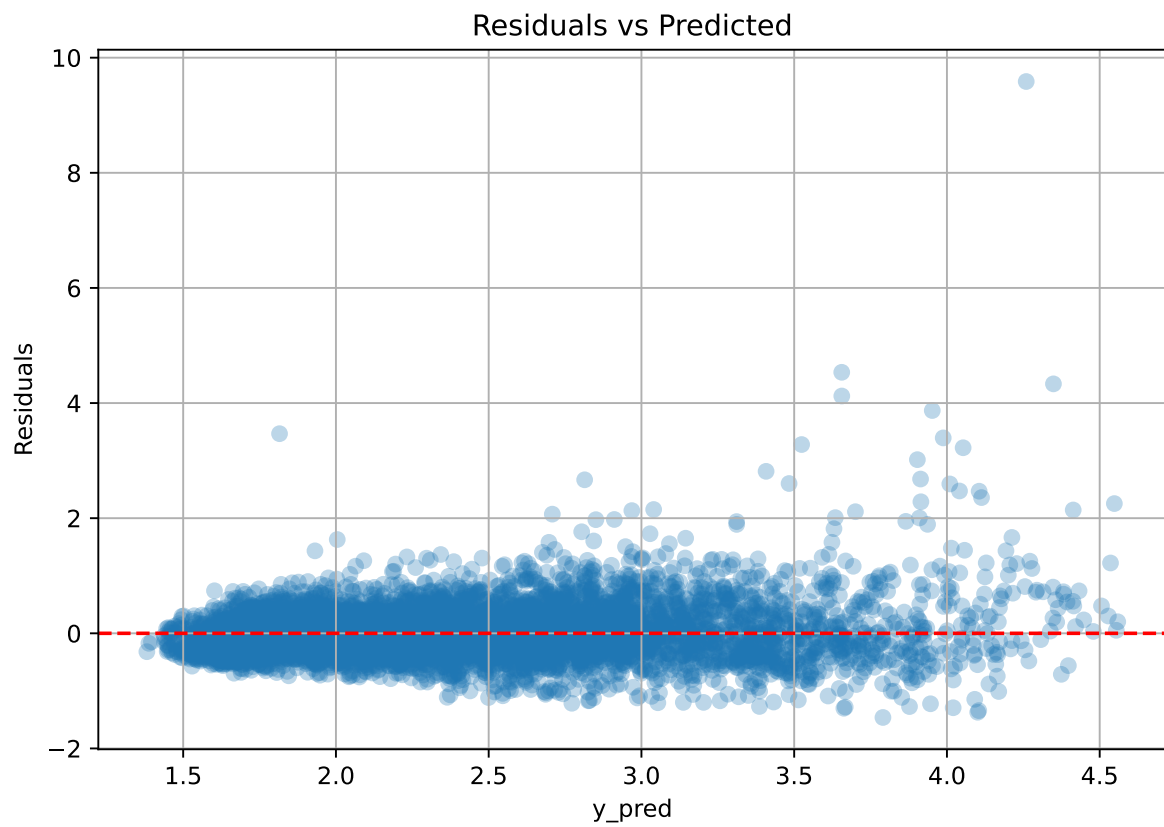
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.04016776861493935, 'reg__num_leaves': 45, 'reg__subsample': 0.5751271479220885, 'reg__colsample_bytree': 0.6301171929625842, 'reg__min_child_samples': 43, 'reg__reg_alpha': 2.961250662699475, 'reg__reg_lambda': 29.73363642169978, 'reg__min_split_gain': 1.2115159195412126, 'reg__path_smooth': 0.19253566123776755, 'reg__min_child_weight': 0.00765631821400957, 'reg__feature_fraction': 0.848576458917341, 'reg__bagging_fraction': 0.644369550207932, 'reg__bagging_freq': 7, 'reg__max_bin': 233, 'reg__min_data_per_group': 179}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.01659538214307651, 'reg__num_leaves': 41, 'reg__subsample': 0.7410014261354534, 'reg__colsample_bytree': 0.7499124207093938, 'reg__min_child_samples': 50, 'reg__reg_alpha': 3.1934343573381847, 'reg__reg_lambda': 30.560056071961593, 'reg__min_split_gain': 1.2859013750073207, 'reg__path_smooth': 1.1966787170337843, 'reg__min_child_weight': 0.04267447052522363, 'reg__feature_fraction': 0.841317492950872, 'reg__bagging_fraction': 0.6423034569537112, 'reg__bagging_freq': 5, 'reg__max_bin': 245, 'reg__min_data_per_group': 172}

3.5.2 Scatter



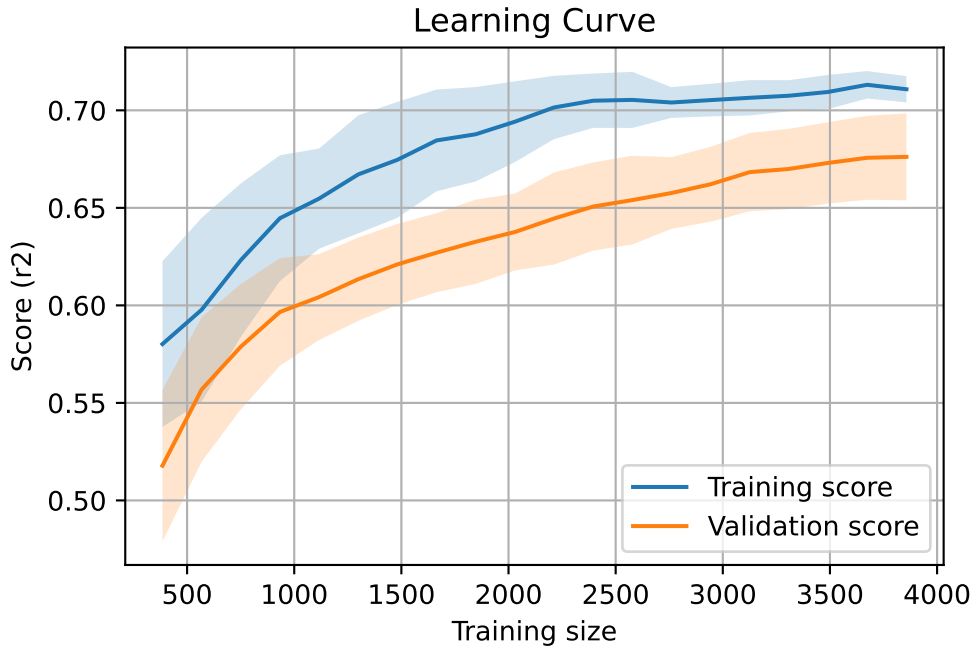
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7874, `Len(training)` : 5512,

`Len(training_real)` : 3859, `Len(test_of_training)` : 1653, `Len(test)` : 2362



3.6 Model : CatBoost

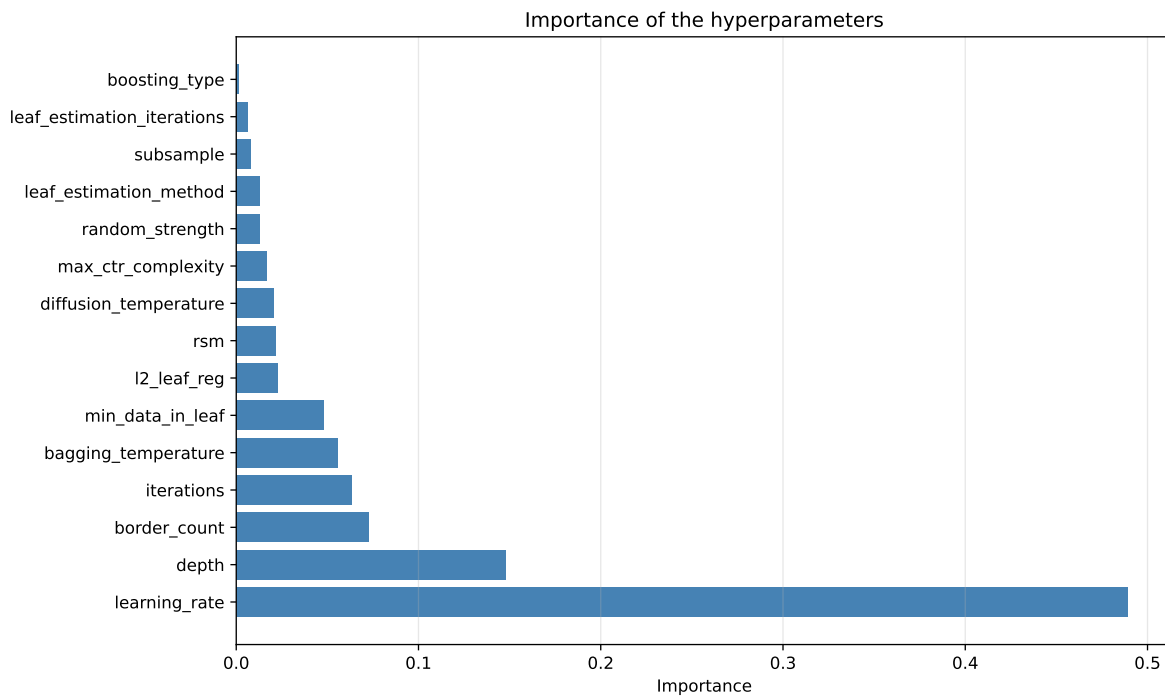
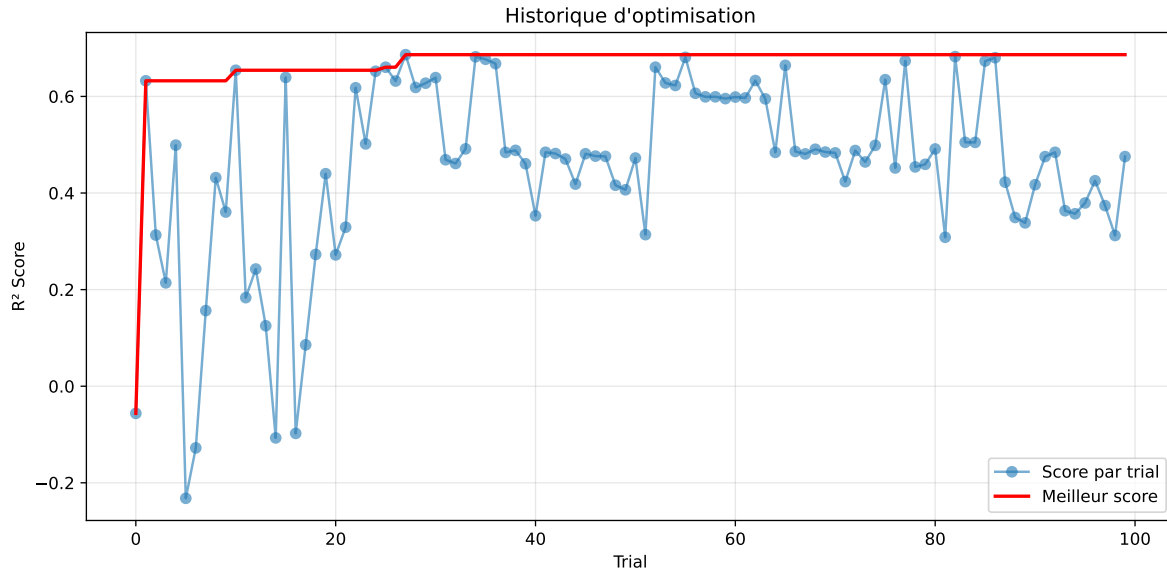
3.6.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.804, min=0.959, max=13.845
 y_{test} : mean=2.342, std=0.779, min=1.062, max=7.779

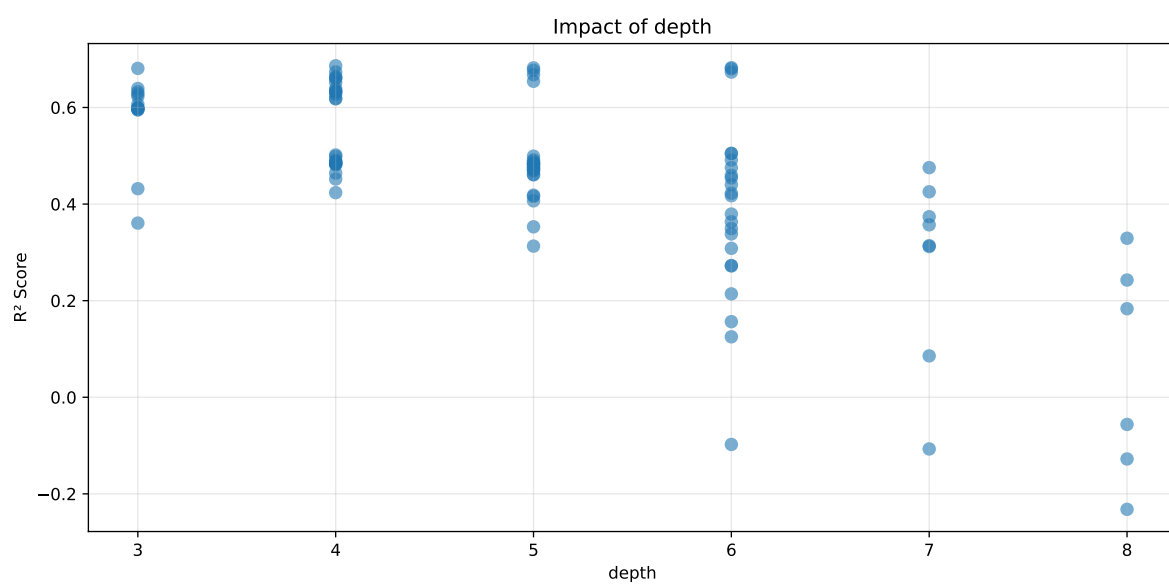
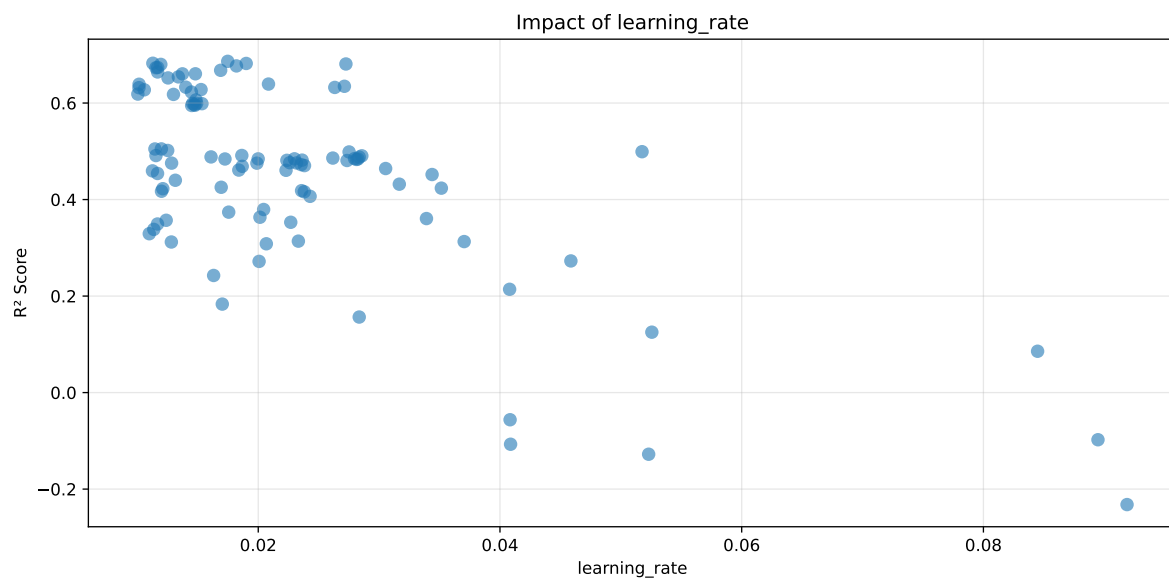
===== TOP Importance FEATURES =====

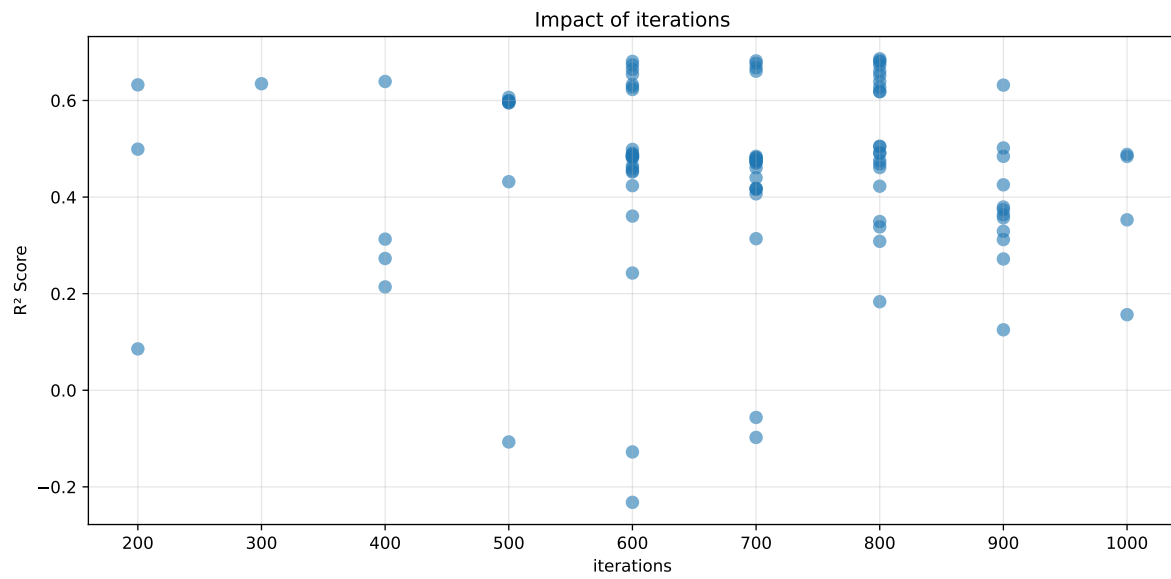
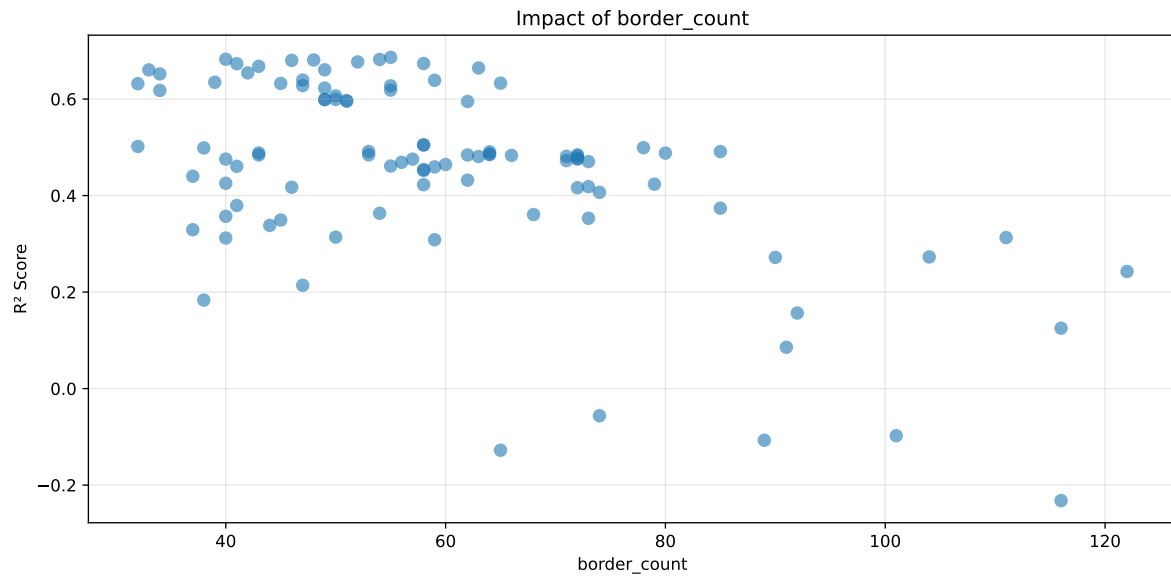
feature importance $r_{umidity}$ 12.981430 $r_{temperature_wl_12h_A_rms}$ 8.775730 r_{CO2} 8.680299 $r_{temperature_wl_24h_D1_energy_rel}$ 8.408500 $r_{temperature}$ 6.386302 $r_{temperature_wl_8h_A_rms}$ 5.027484 r_{NH3} 4.726424 $r_{umidity_wl_24h_A_energy}$ 3.015428 $r_{temperature_wl_12h_A_energy}$ 2.783213 $r_{temperature_wl_8h_D1_energy}$ 2.574581 r_{THI} 2.368242 $r_{NH3_wl_24h_D3_energy}$ 2.232420 $r_{umidity_wl_12h_D2_energy_rel}$ 2.184631 $r_{umidity_wl_8h_D1_energy_rel}$ 1.907960 $r_{umidity_wl_12h_A_rms}$ 1.815019 $r_{NH3_wl_12h_A_energy}$ 1.584535 $r_{NH3_wl_24h_A_energy}$ 1.527198 $r_{umidity_wl_24h_D2_energy_rel}$ 1.488613 $r_{NH3_wl_24h_D2_energy}$ 1.484333 $r_{temperature_wl_12h_D2_energy}$ 1.426537 $r_{NH3_wl_24h_D3_energy_rel}$ 1.406853 $r_{temperature_wl_8h_A_energy}$ 1.334180 $r_{NH3_wl_24h_D1_energy}$ 1.314652 $r_{temperature_wl_24h_D3_energy_rel}$ 1.312612 $r_{NH3_wl_24h_A_rms}$ 1.240140 $r_{NH3_wl_12h_D2_energy}$ 1.160163 $r_{NH3_wl_8h_A_rms}$ 1.138906 $r_{umidity_wl_24h_D1_energy}$ 1.012007 $r_{umidity_wl_12h_D1_energy}$ 0.967209 $r_{umidity_wl_24h_D3_energy_rel}$ 0.966871 $r_{NH3_wl_12h_A_rms}$ 0.963818 $r_{NH3_wl_8h_A_en-}$

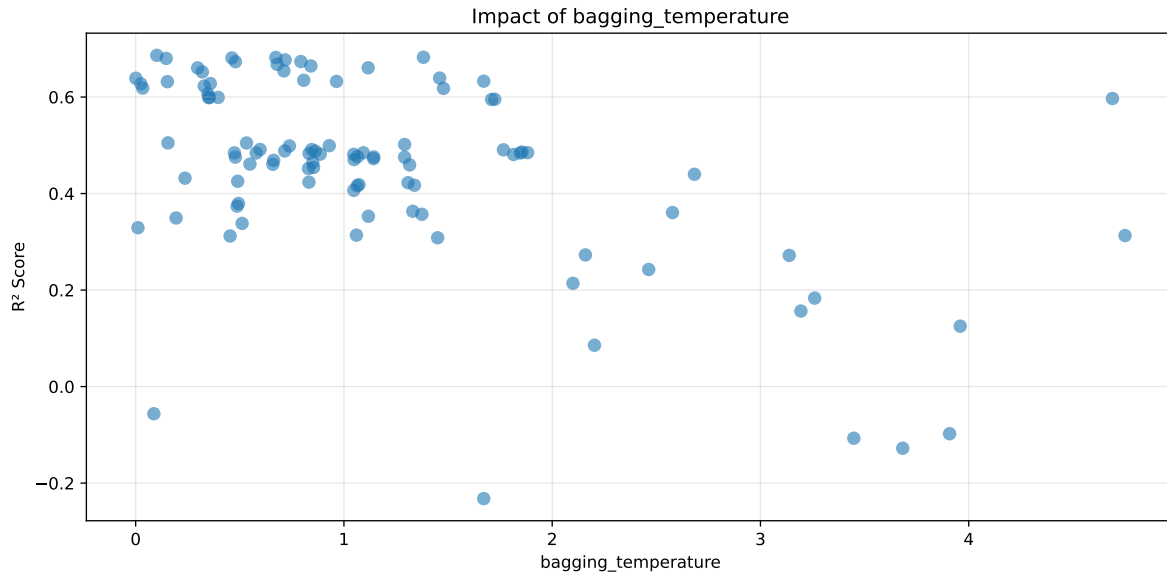
ergy 0.923749 r_NH3_wl_12h_D2_energy_rel 0.918162 r_umidity_wl_8h_A_rms
 0.784138 r_NH3_wl_8h_D1_energy 0.690201 r_NH3_wl_12h_D1_energy 0.667902
 r_NH3_wl_24h_A_energy_rel 0.608927 r_umidity_wl_24h_A_rms 0.495725 r_NH3_wl_8h_D1_en-
 ergy_rel 0.402949 r_NH3_wl_12h_D1_energy_rel 0.311959



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'border_count', 'iterations', 'bagging_temperature']

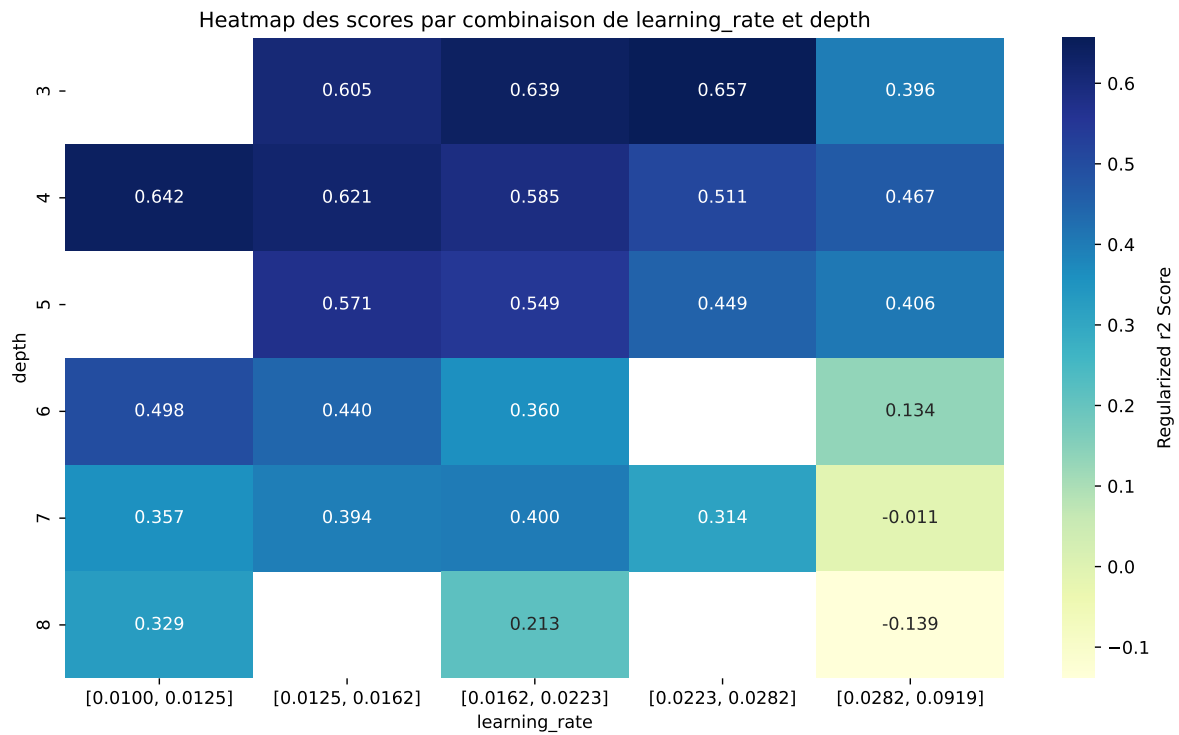






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7184	0.6864	0.6845	0.023	0.6864	0.6947	0.0082	1.012	1.0466
2	0.7164	0.6825	0.6792	0.0211	0.6825	0.693	0.0105	1.0154	1.0498
3	0.7153	0.682	0.679	0.0218	0.682	0.6966	0.0146	1.0214	1.0488
4	0.703	0.6809	0.676	0.022	0.6809	0.6838	0.0028	1.0042	1.0323
5	0.714	0.6801	0.6776	0.0214	0.6801	0.6908	0.0107	1.0157	1.0499
6	0.7083	0.6769	0.6757	0.0219	0.6769	0.69	0.013	1.0193	1.0463
7	0.7022	0.6736	0.6665	0.0225	0.6736	0.6714	-0.0022	0.9968	1.0425
8	0.706	0.6732	0.6707	0.0215	0.6732	0.6841	0.0109	1.0162	1.0488
9	0.6935	0.6678	0.6646	0.0208	0.6678	0.6789	0.0111	1.0166	1.0384
10	0.6913	0.6644	0.658	0.0205	0.6644	0.666	0.0016	1.0024	1.0405
---	---	---	---	---	---	---	---	---	---
100	0.9907	0.7869	0.781	0.0321	-0.2321	0.8005	0.0135	1.0172	1.259

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.017475534804300013, 'reg__l2_leaf_reg': 29.628411281039007, 'reg__bagging_temperature': 0.10156376215057622, 'reg__random_strength': 3.5749062269458145, 'reg__subsample': 0.5550850502298493, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5243111202215142, 'reg__border_count': 55, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5500.989892453724}

Rank 2: {'reg__iterations': 800, 'reg__depth': 6, 'reg__learning_rate': 0.011278780369516326, 'reg__l2_leaf_reg': 25.747173308223864, 'reg__bagging_temperature': 1.3818785067544546, 'reg__random_strength': 2.252012525240695, 'reg__subsample': 0.5758448802860694, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6330174492097316, 'reg__border_count': 40, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6116.075043088216}

Rank 3: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.019003863976678803, 'reg__l2_leaf_reg': 35.382844053707224, 'reg__bagging_temperature': 0.6729902555333525, 'reg__random_strength': 3.5750207843665307, 'reg__subsample': 0.5845056647220288, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5475860781610288, 'reg__border_count': 54, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7634.303085682768}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.027253806427991176, 'reg__l2_leaf_reg': 39.59565247640794, 'reg__bagging_temperature': 0.4621952356425504, 'reg__random_strength': 3.2450360198884964, 'reg__subsample': 0.5315993213461342, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 3, 'reg__rsm':

0.5951353679990052, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8701.115744330038}

Rank 5: {'reg__iterations': 800, 'reg__depth': 6, 'reg__learning_rate': 0.01193826826586168, 'reg__l2_leaf_reg': 29.85320972684287, 'reg__bagging_temperature': 0.14662506978839923, 'reg__random_strength': 3.029392151567788, 'reg__subsample': 0.5541894389737025, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6094498303151038, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 4643.765428988459}

Rank 6: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.01819700973545884, 'reg__l2_leaf_reg': 35.216242863527775, 'reg__bagging_temperature': 0.7175834668193981, 'reg__random_strength': 3.5699037700670897, 'reg__subsample': 0.5063828517400284, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5484281622205128, 'reg__border_count': 52, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7879.21145553885}

Rank 7: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.011680921090025286, 'reg__l2_leaf_reg': 45.16723930206787, 'reg__bagging_temperature': 0.7932599958144981, 'reg__random_strength': 0.15932736765827996, 'reg__subsample': 0.6761759522734013, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.633370963310228, 'reg__border_count': 58, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6294.584464238095}

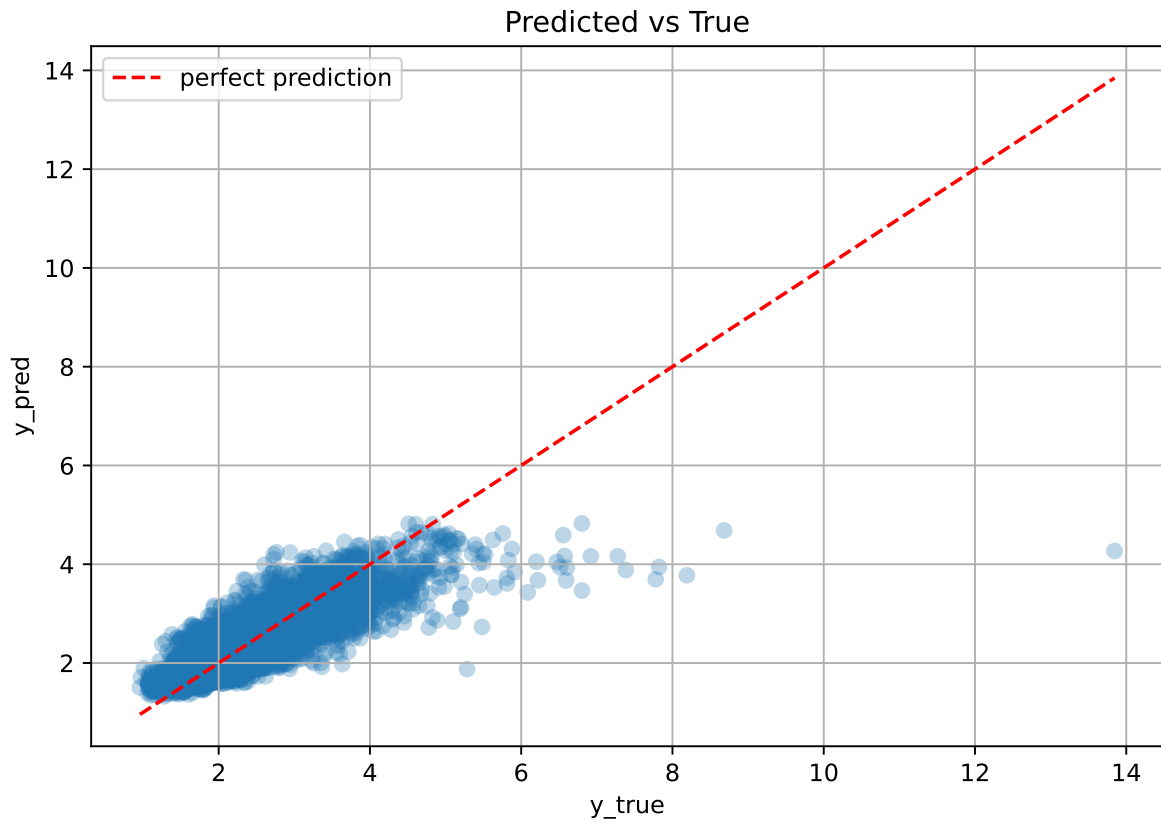
Rank 8: {'reg__iterations': 800, 'reg__depth': 6, 'reg__learning_rate': 0.011537361957347529, 'reg__l2_leaf_reg': 29.4372620959897, 'reg__bagging_temperature': 0.47915926792445623, 'reg__random_strength': 3.00415670883673, 'reg__subsample': 0.5524310741993254, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6125510916547052, 'reg__border_count': 41, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2897.185357936678}

Rank 9: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.01688973495956959, 'reg__l2_leaf_reg': 35.76915165333823, 'reg__bagging_temperature': 0.6792222559905394, 'reg__random_strength': 3.519305272362253, 'reg__subsample': 0.6119297404703443, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5492344557863995, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 3530.0869356902517}

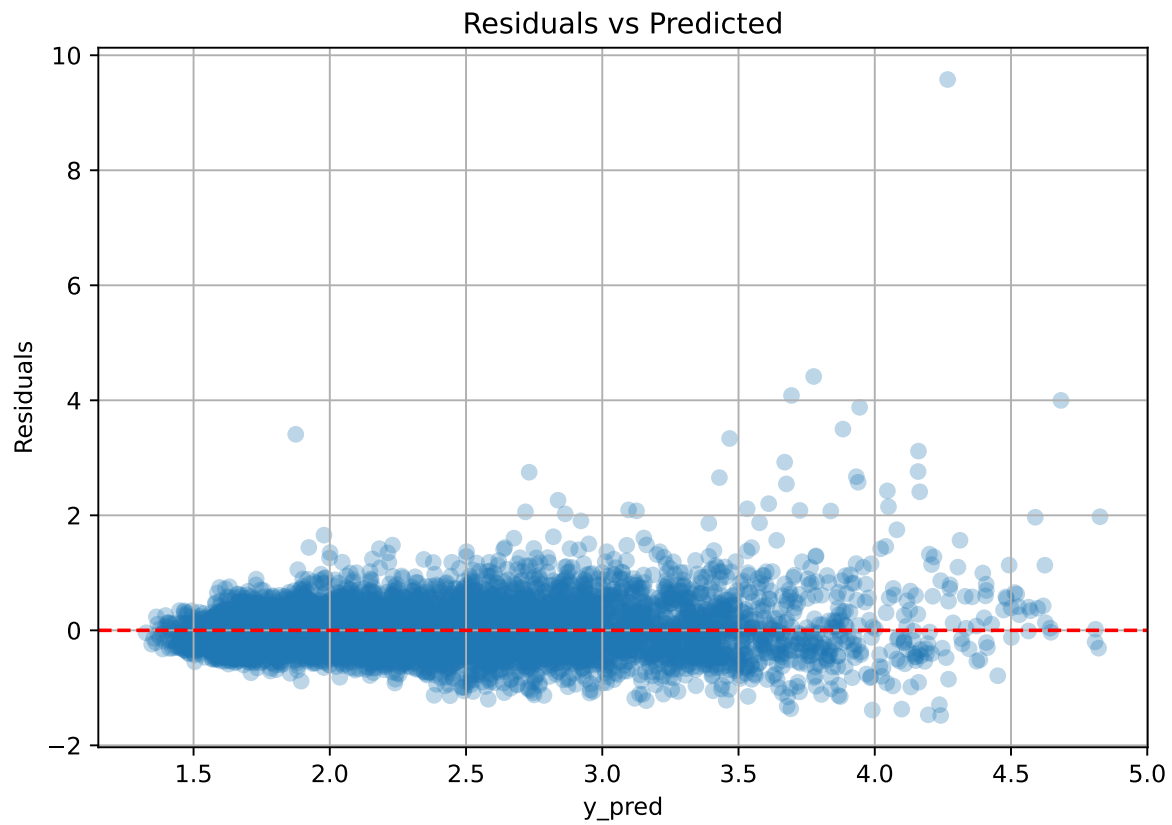
Rank 10: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.011672588304867745, 'reg__l2_leaf_reg': 42.85826973952786, 'reg__bagging_temperature': 0.841271124127085,


```
'reg__random_strength': 0.056376422340044474, 'reg__subsample': 0.574556847375345,  
'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':  
0.5684222736159873, 'reg__border_count': 63, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_tempera-  
ture': 7008.186572543561}
```

3.6.2 Scatter



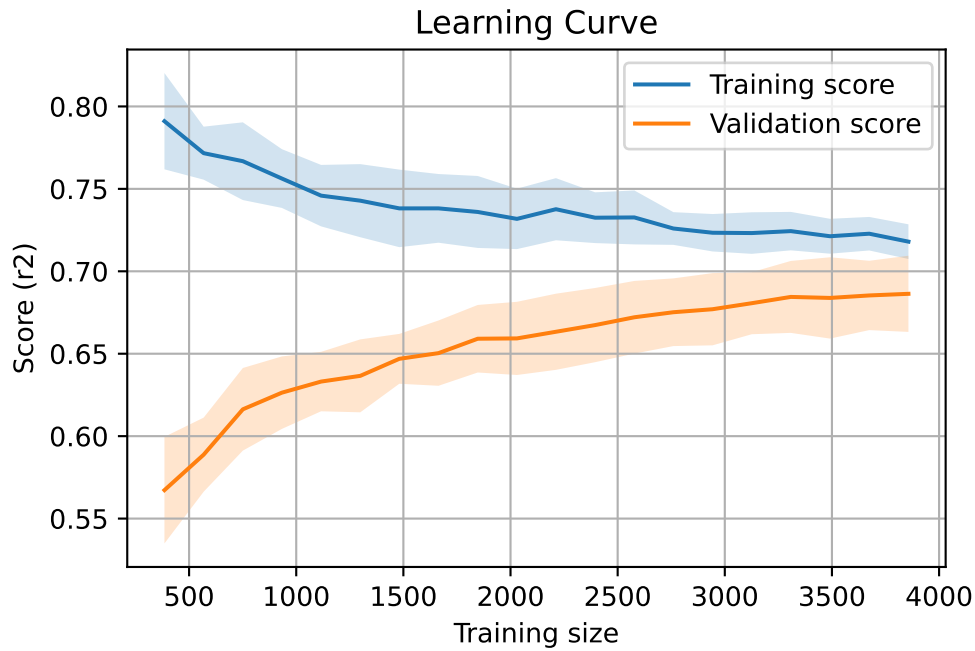
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7874, `Len(training)` : 5512,

`Len(training_real)` : 3859, `Len(test_of_training)` : 1653, `Len(test)` : 2362



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.565	0.556	0.02	0.557	0.56	0.004	1.008	1.016	nan
LGBM	0.711	0.678	0.023	0.673	0.689	0.011	1.017	1.049	0.678
CatBoost	0.718	0.686	0.023	0.684	0.695	0.008	1.012	1.047	0.6864

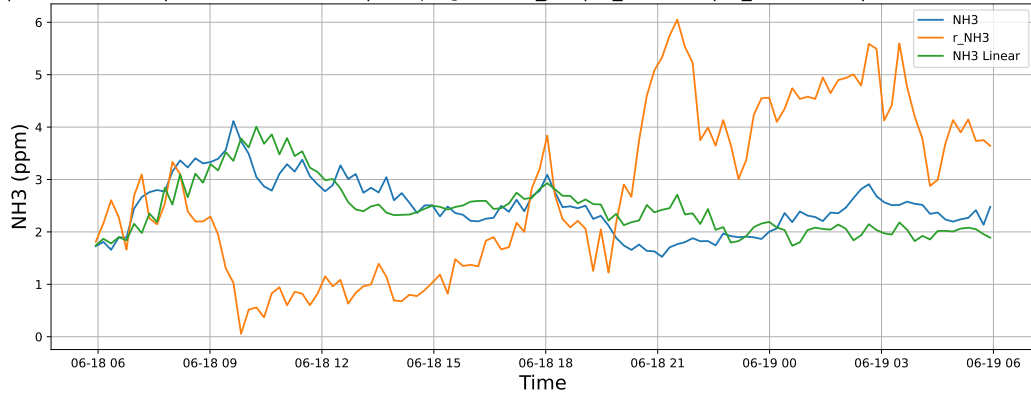
4 Plot

4.1 Standalone plots

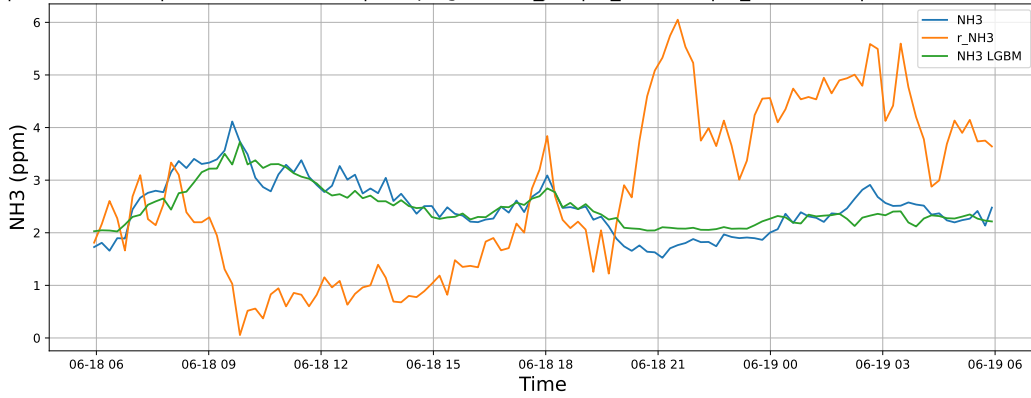
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

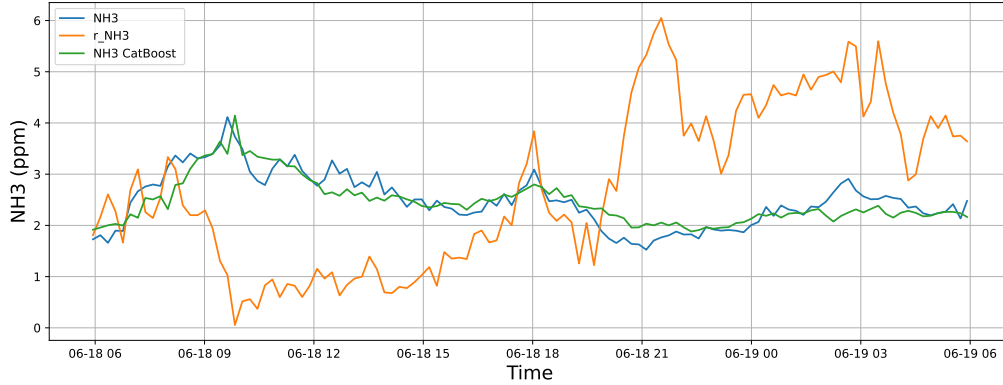
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

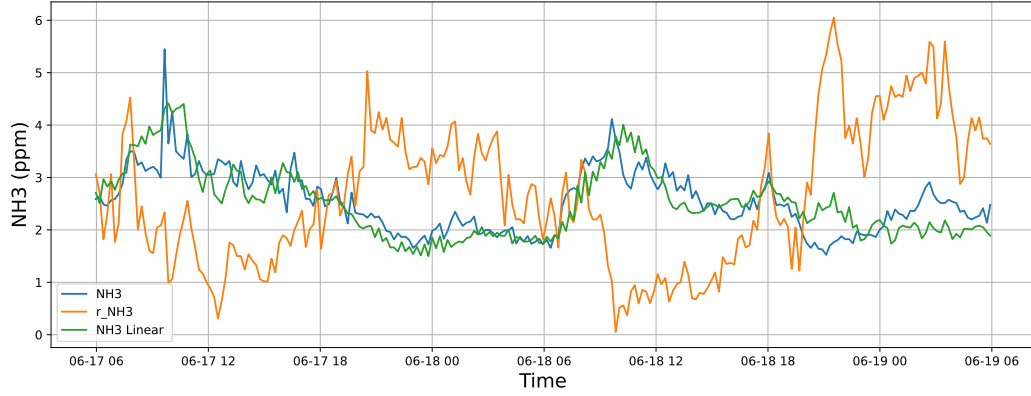


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

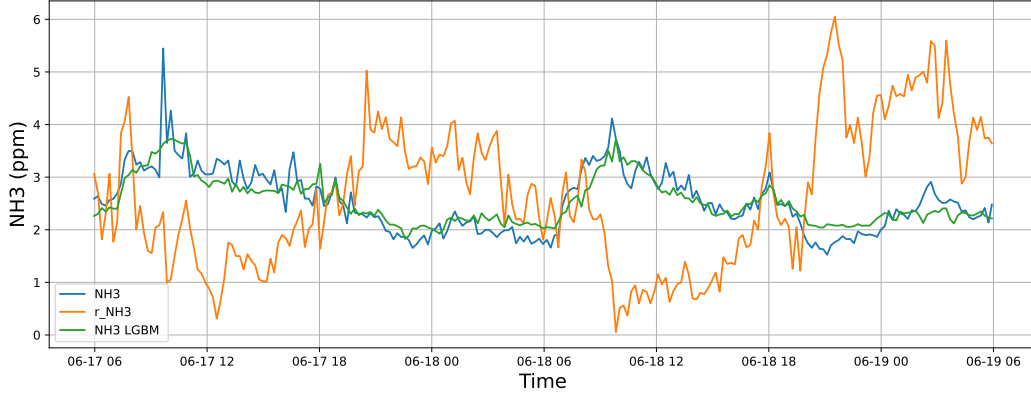


4.1.1.2 Time window : 48

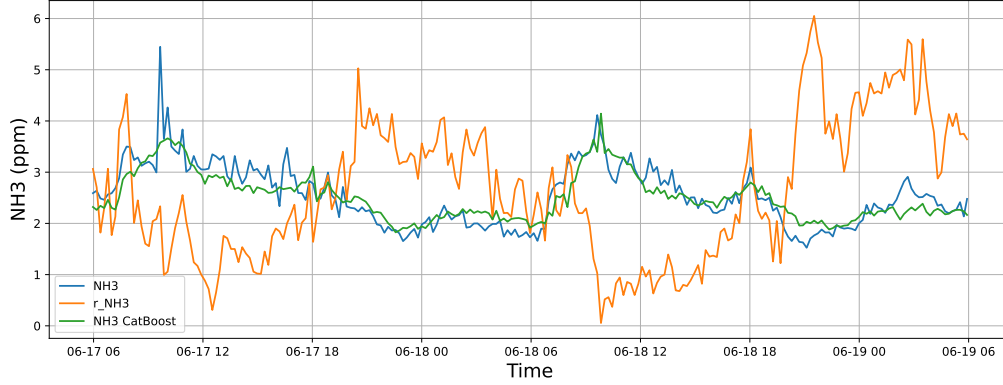
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

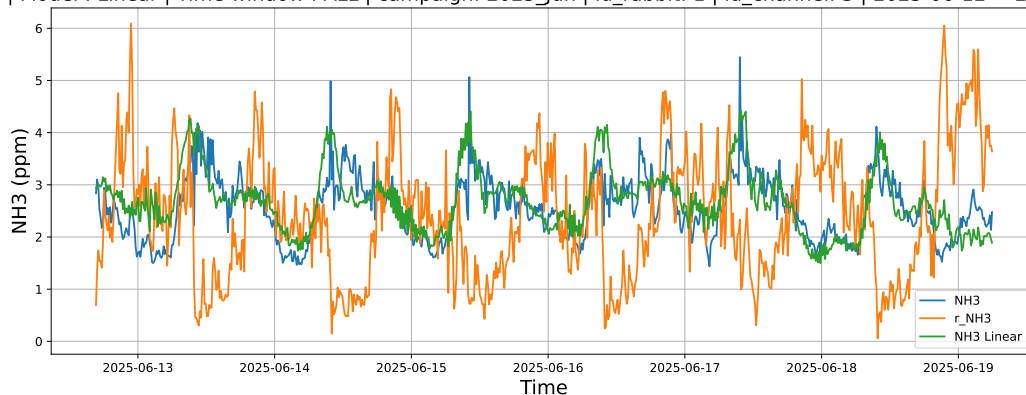


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

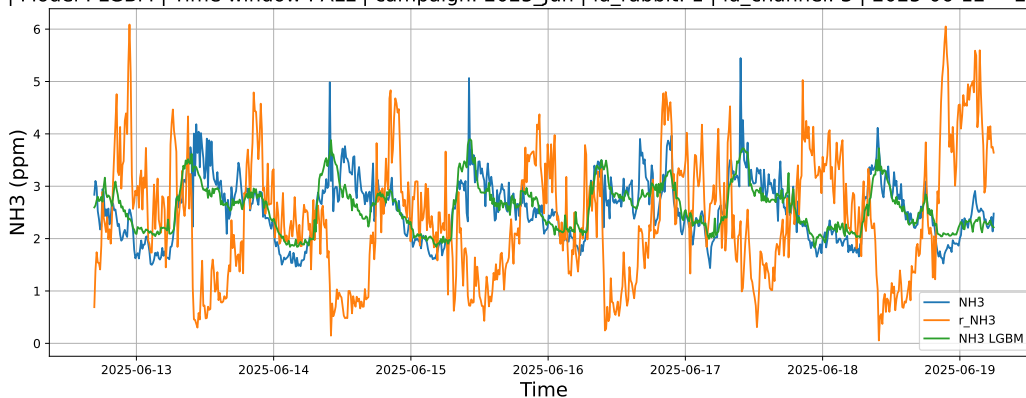


4.1.1.3 Time window : ALL

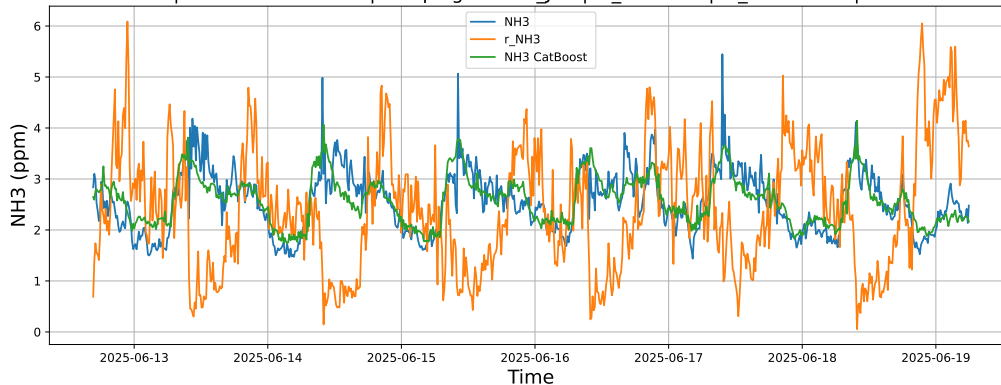
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



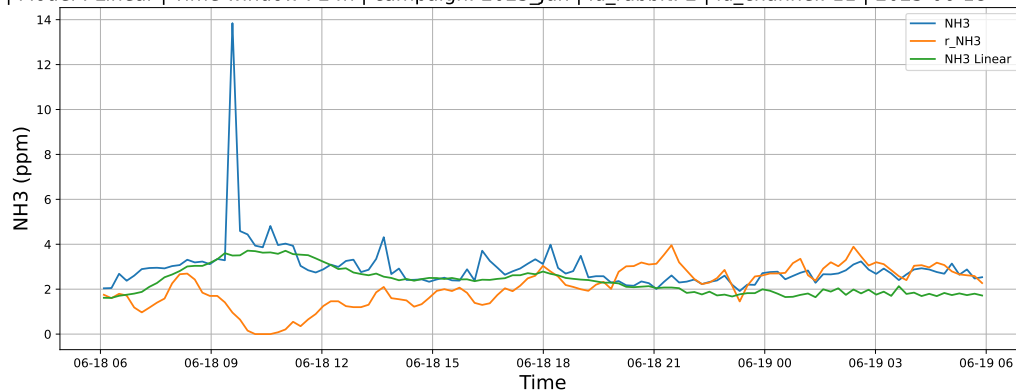
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



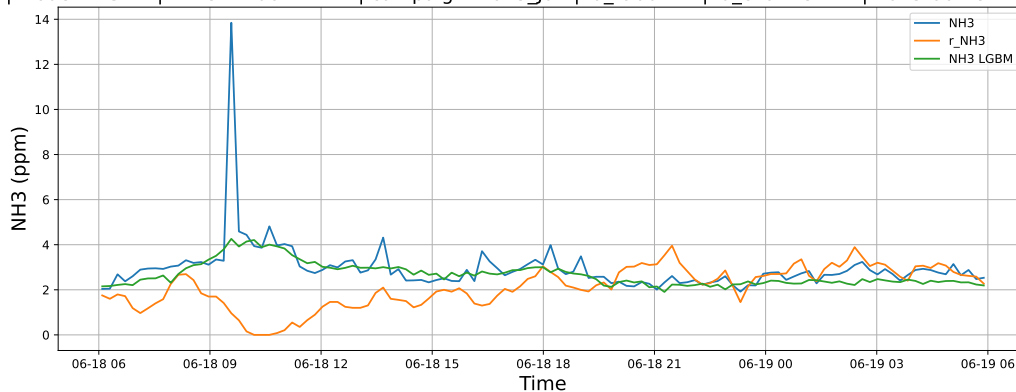
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

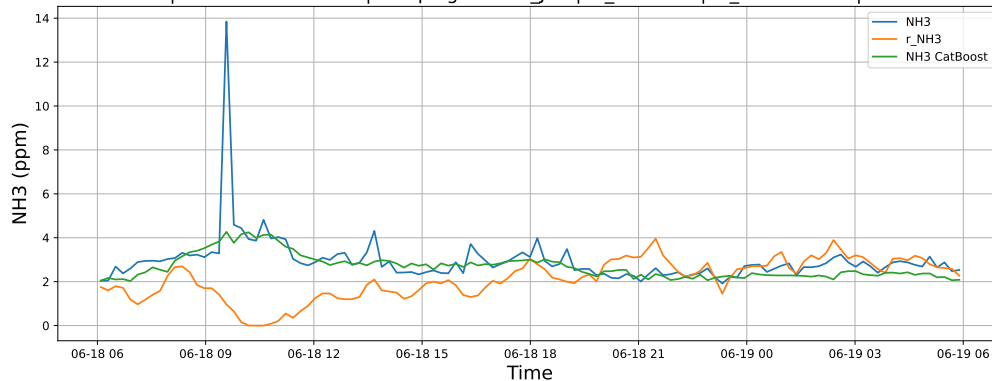
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

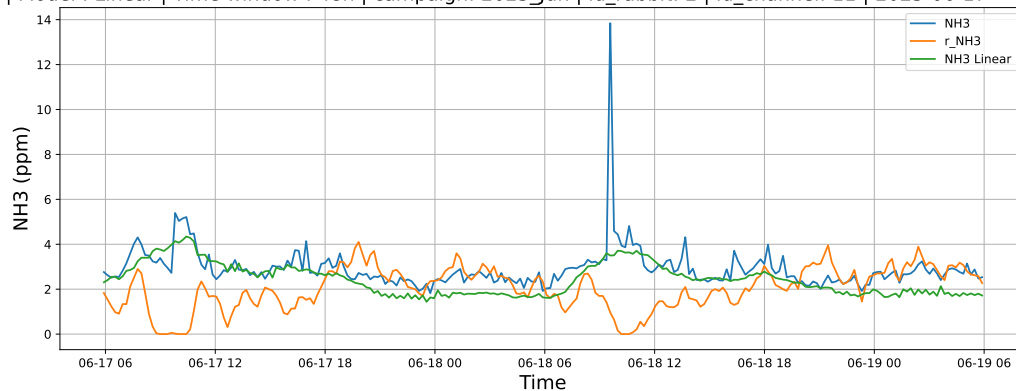


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

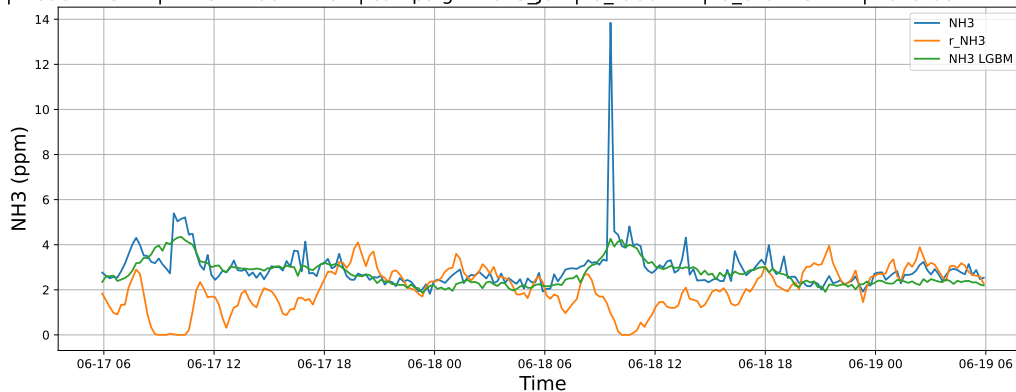


4.1.2.2 Time window : 48

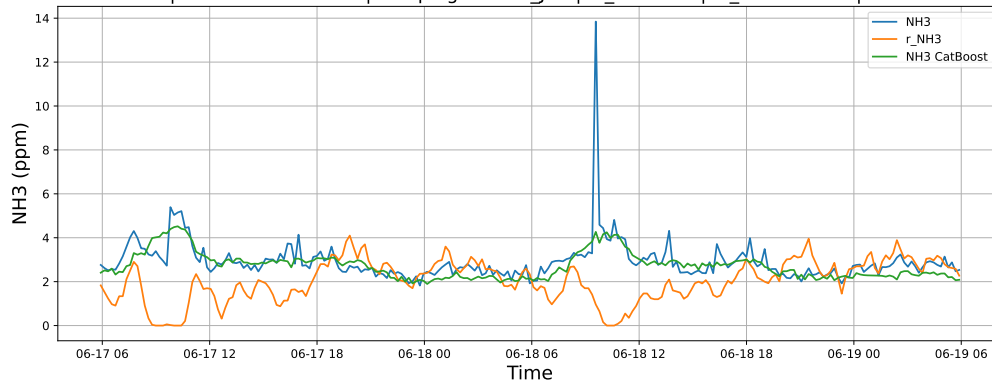
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

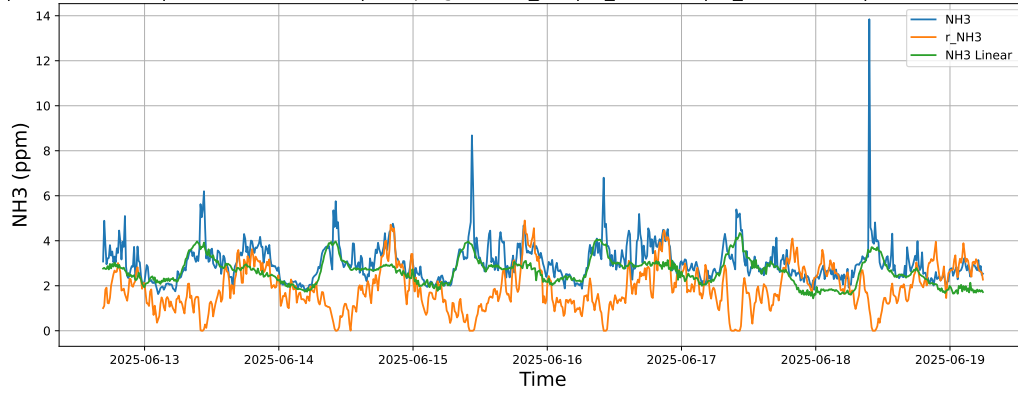


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

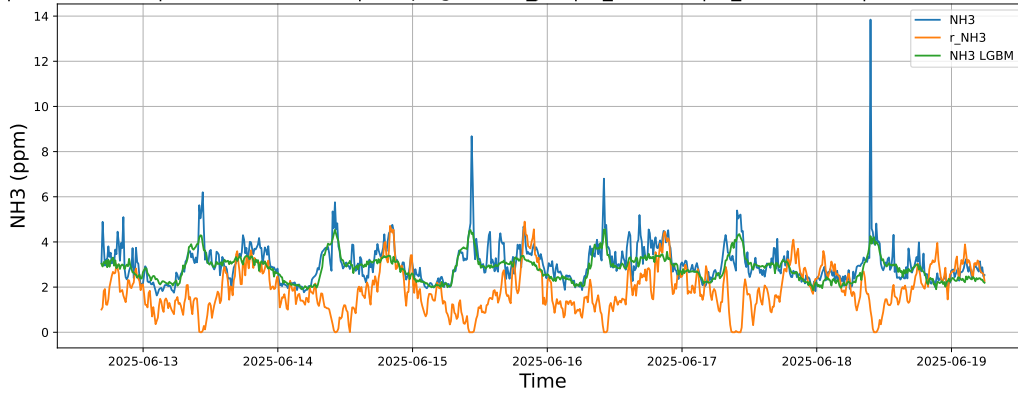


4.1.2.3 Time window : ALL

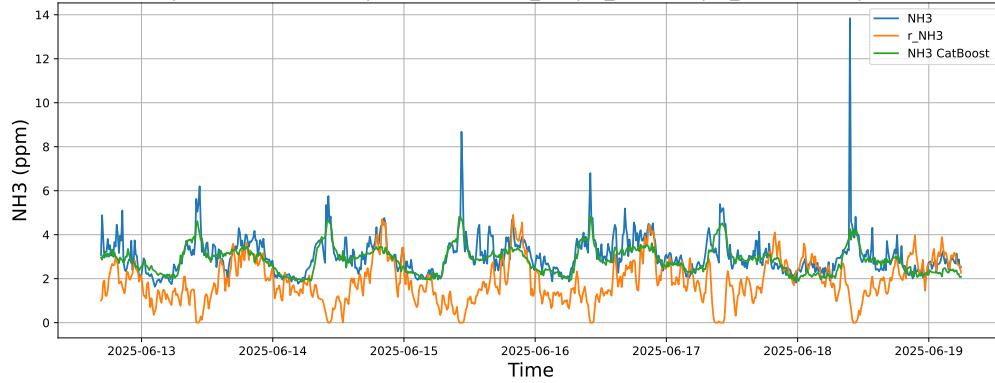
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



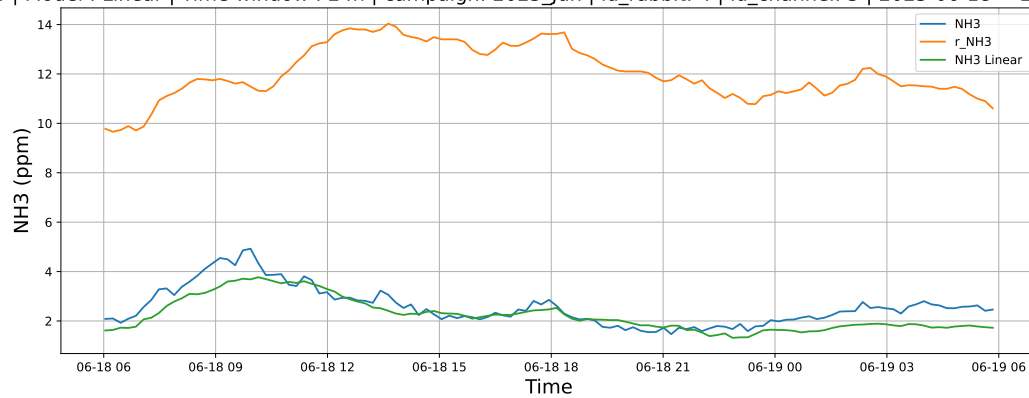
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



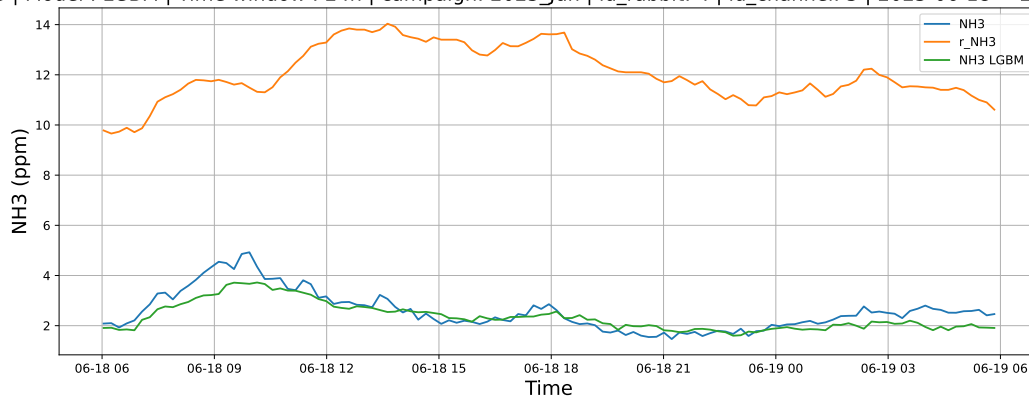
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

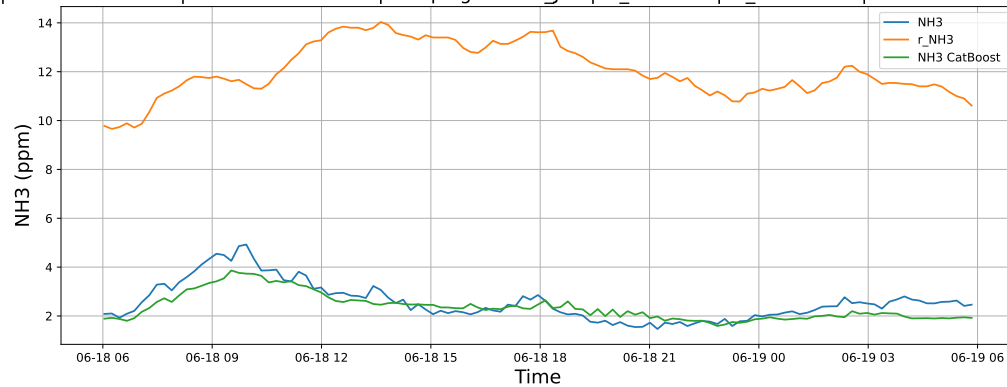
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

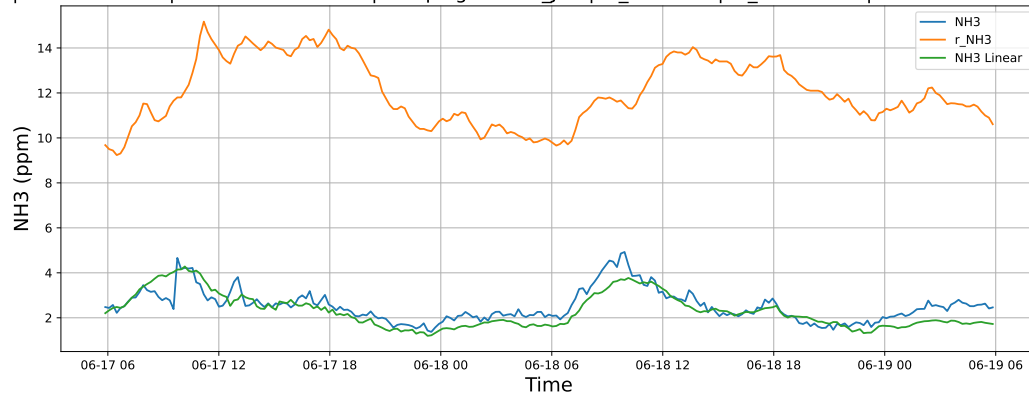


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

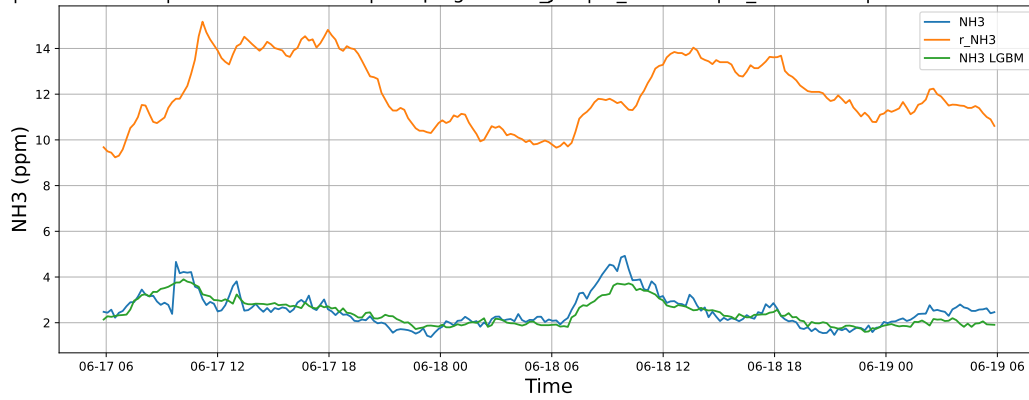


4.1.3.2 Time window : 48

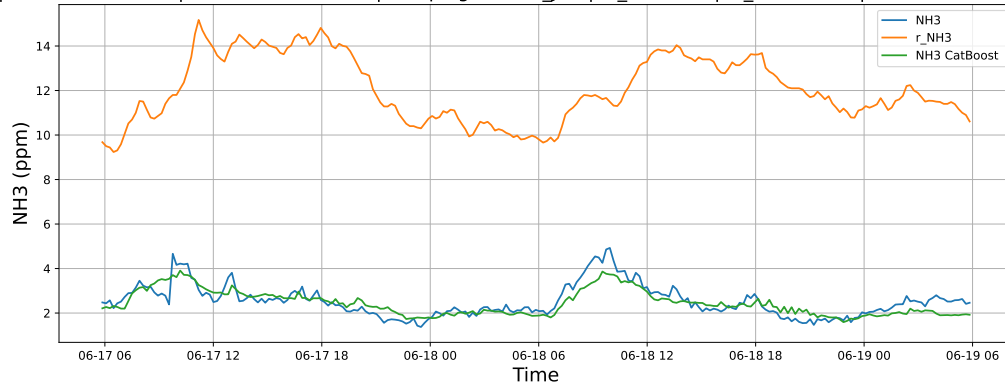
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

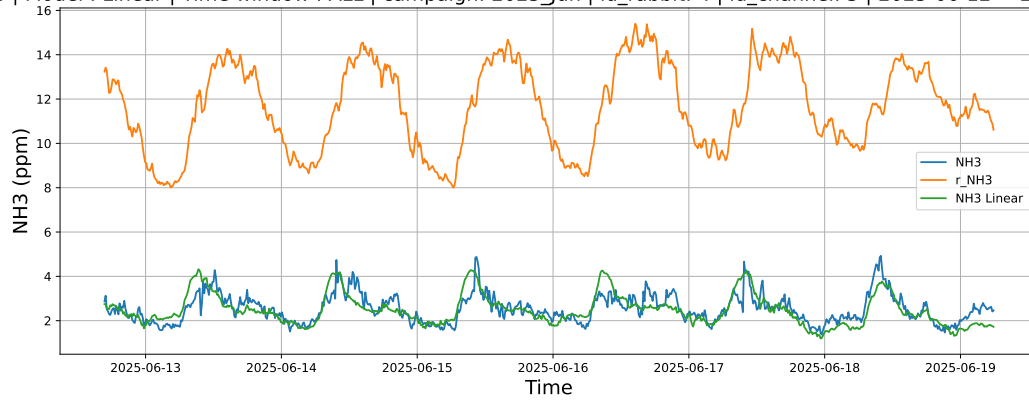


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

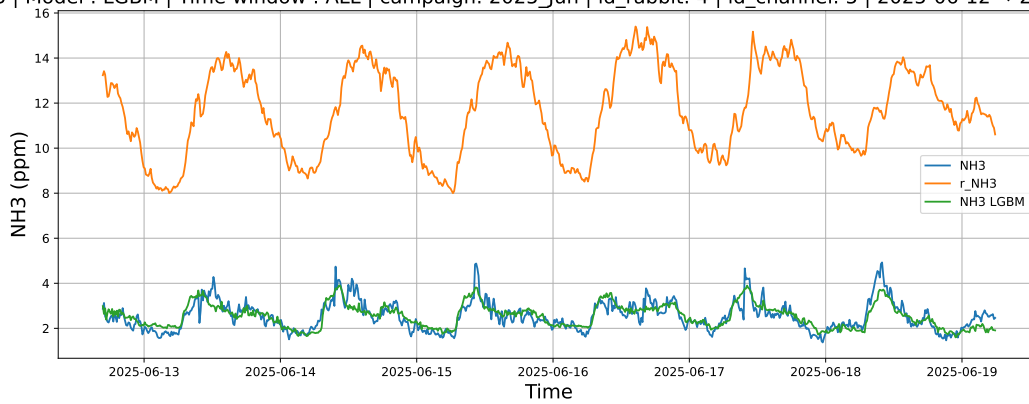


4.1.3.3 Time window : ALL

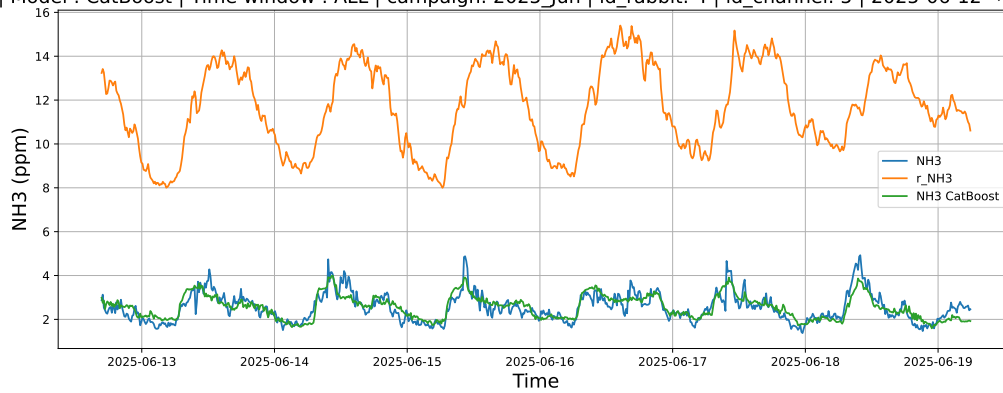
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



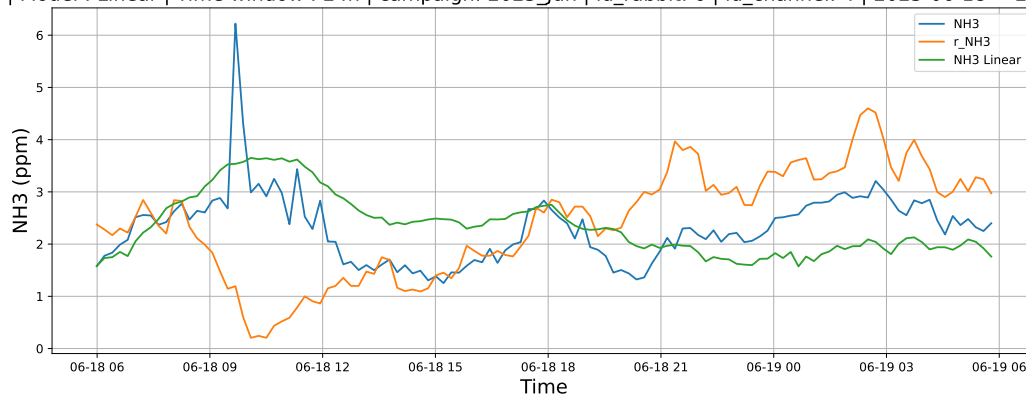
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



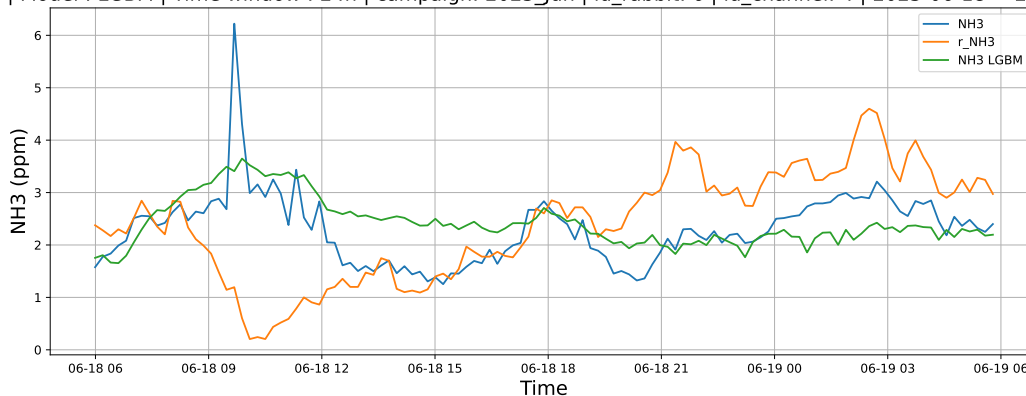
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

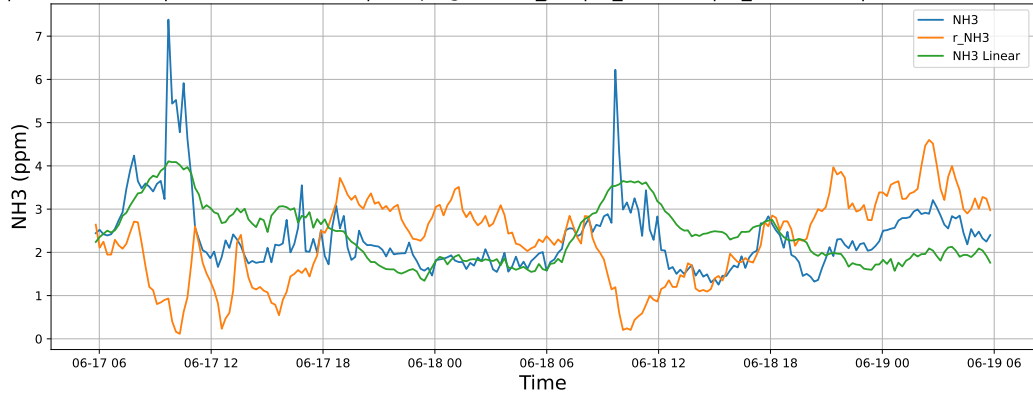


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

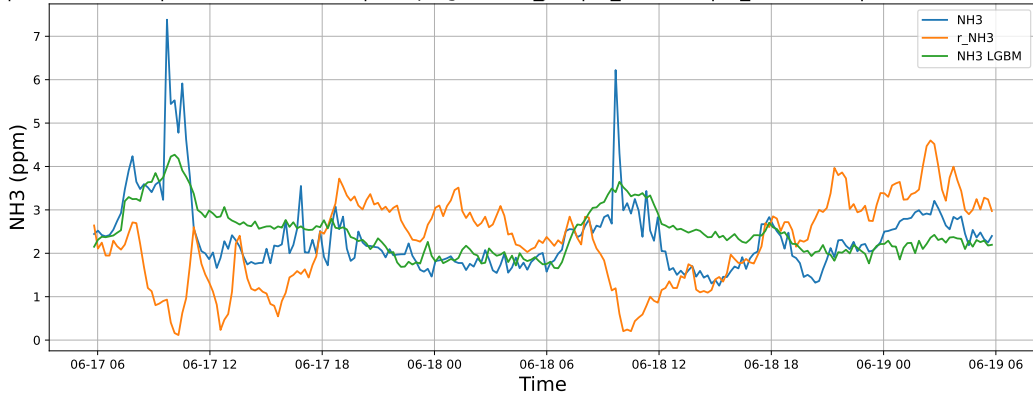


4.1.4.2 Time window : 48

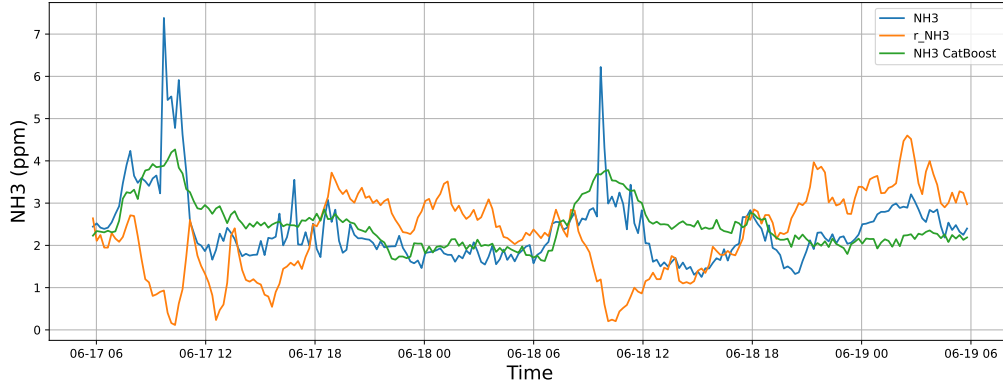
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

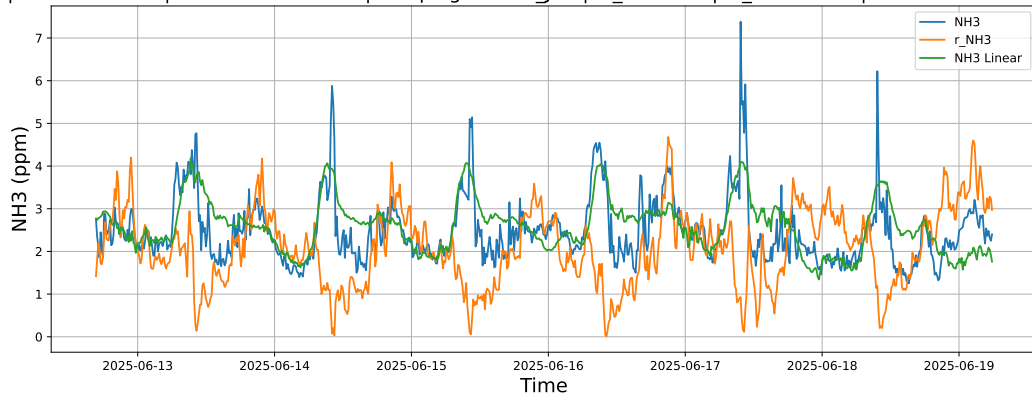


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

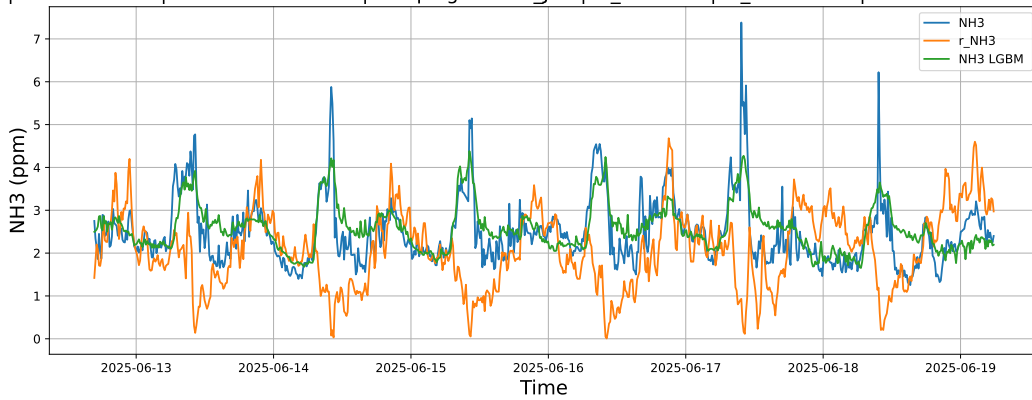


4.1.4.3 Time window : ALL

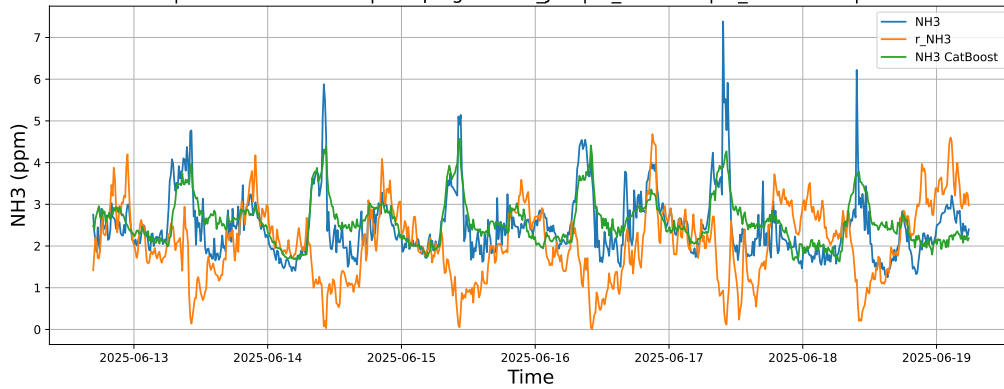
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



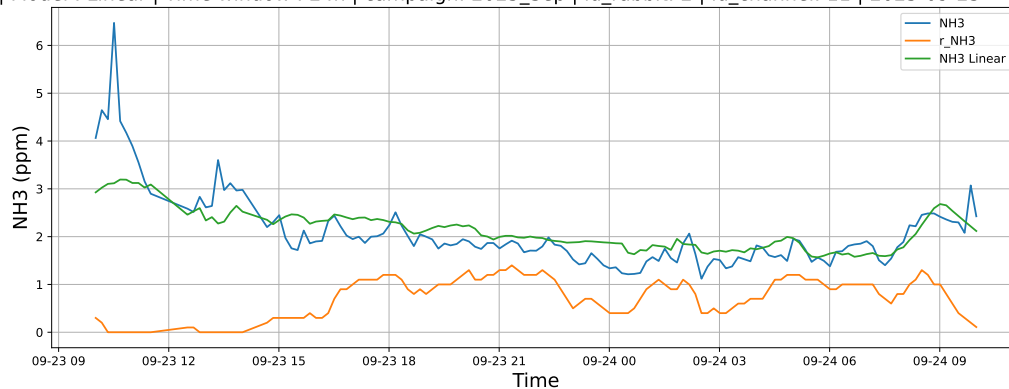
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



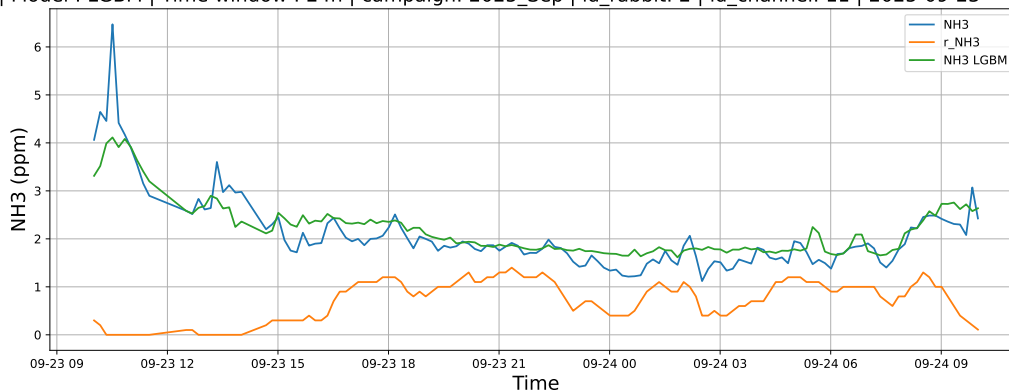
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

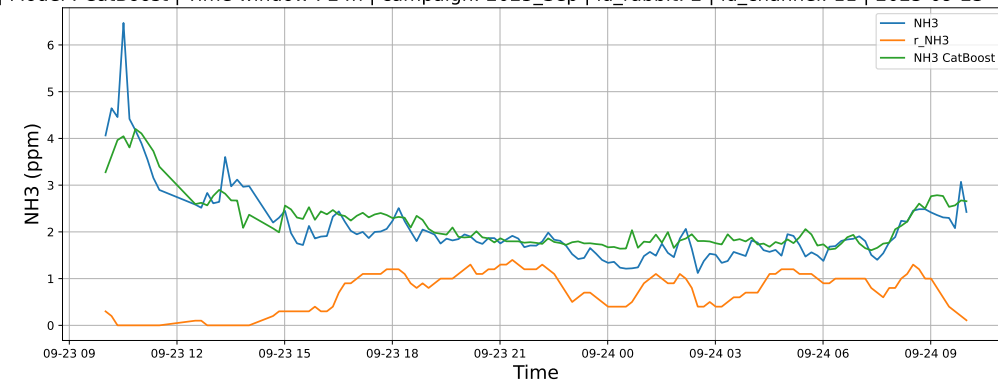
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

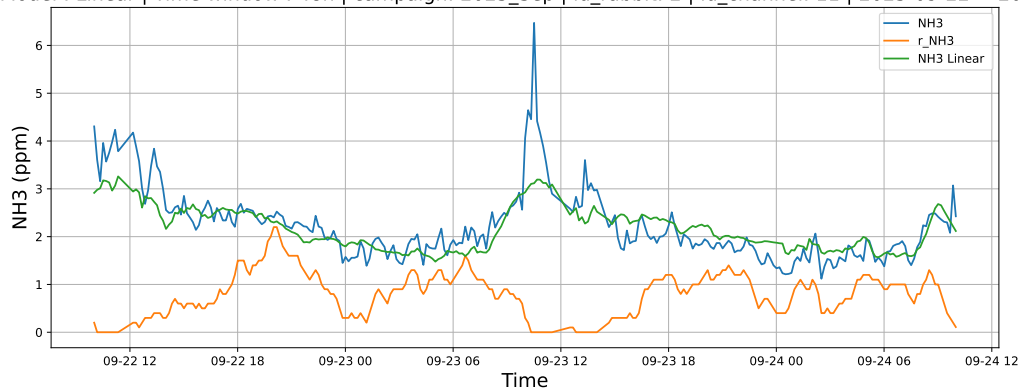


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

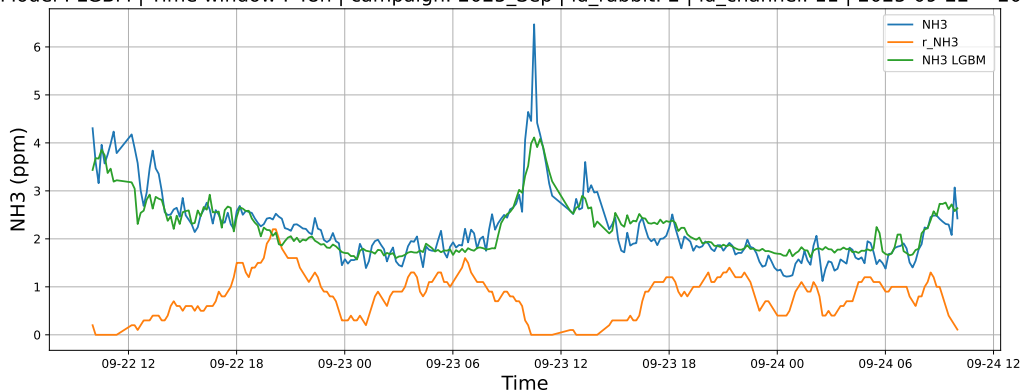


4.1.5.2 Time window : 48

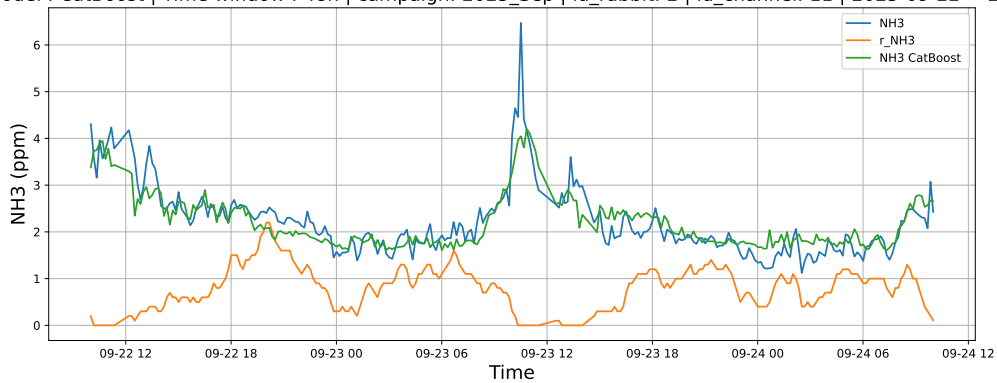
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

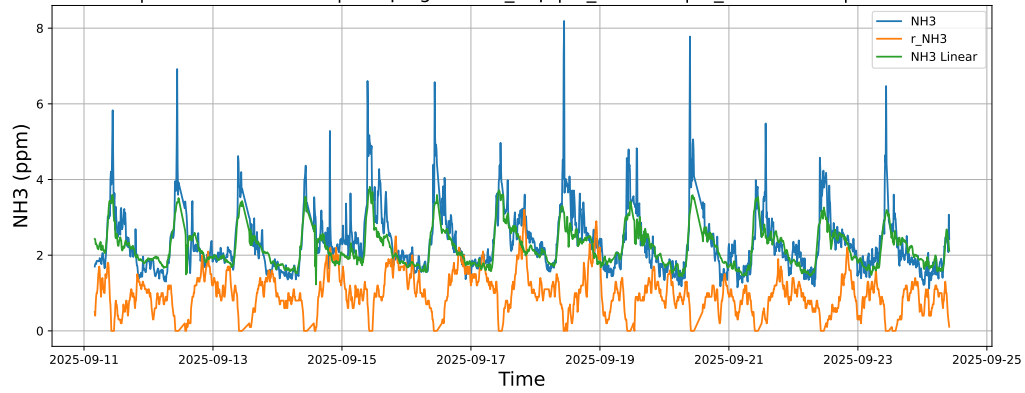


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

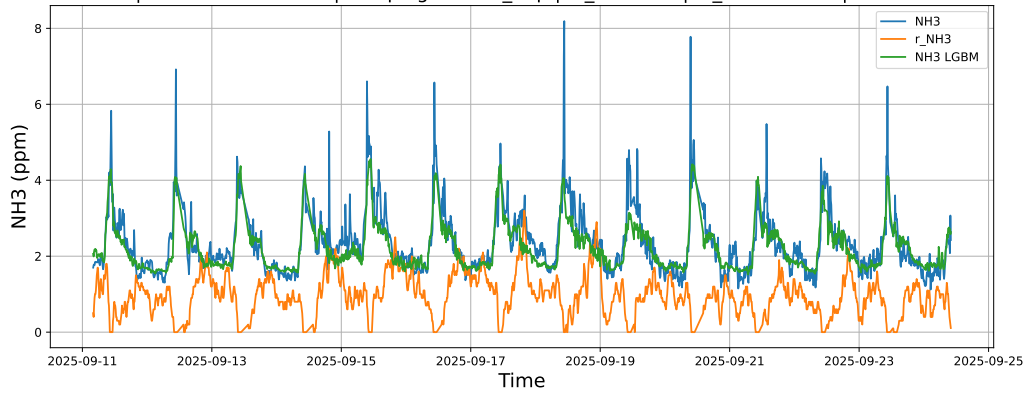


4.1.5.3 Time window : ALL

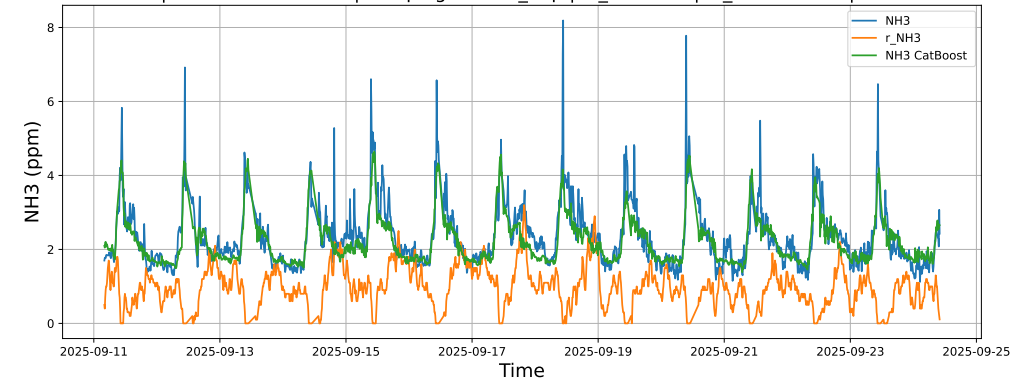
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



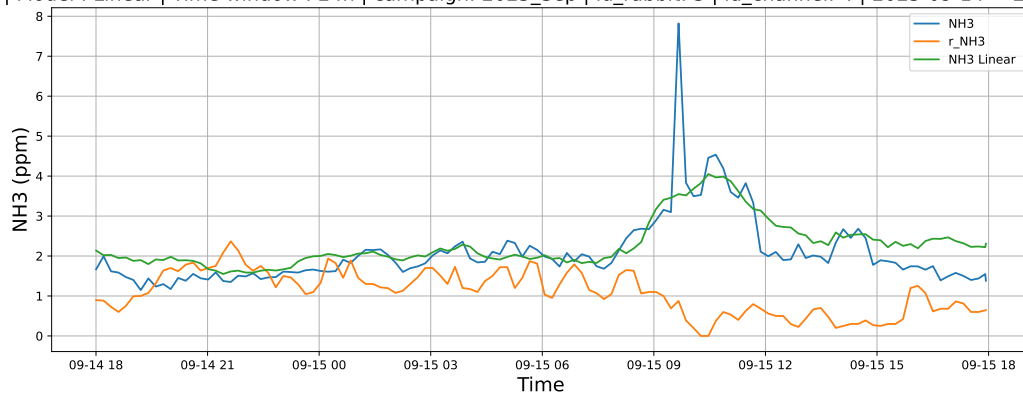
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



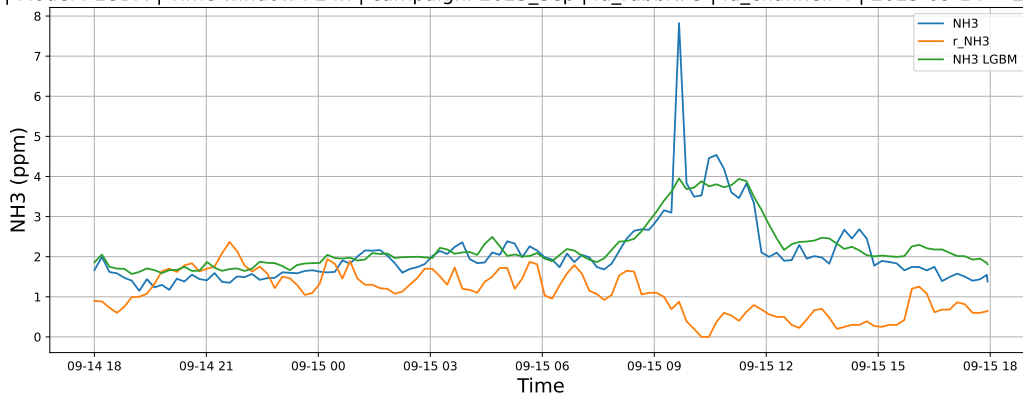
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

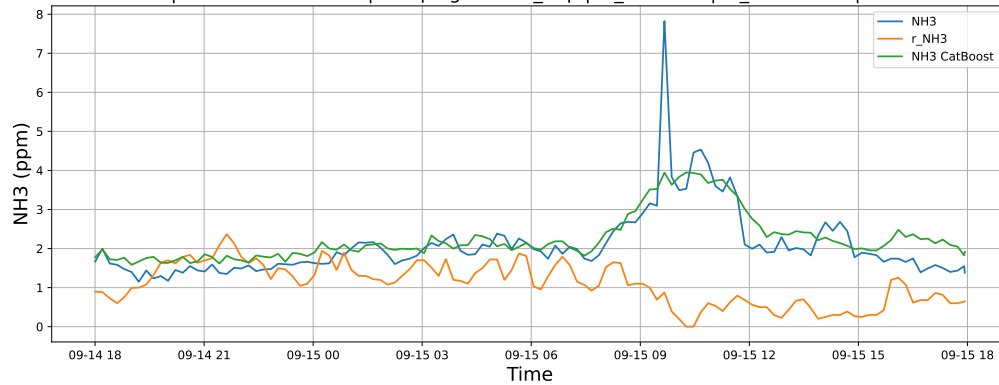
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

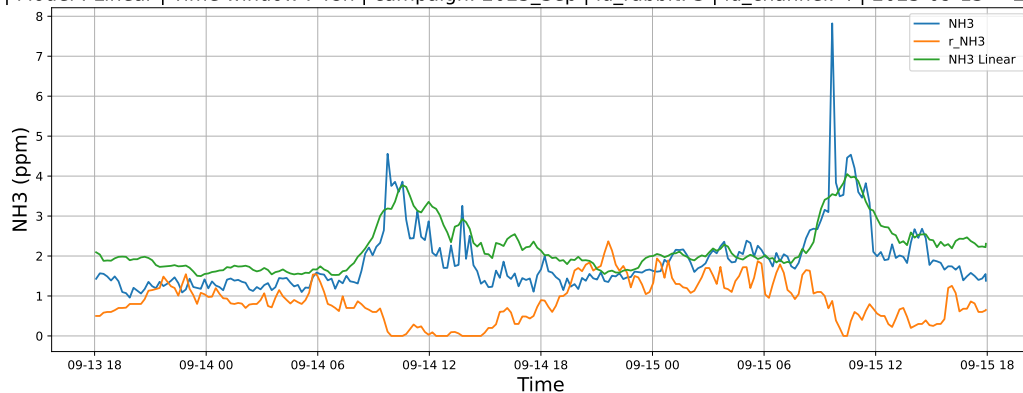


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

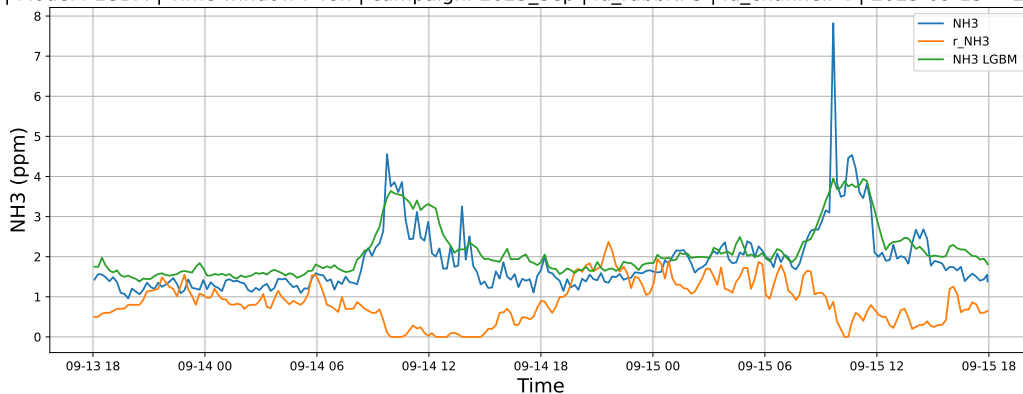


4.1.6.2 Time window : 48

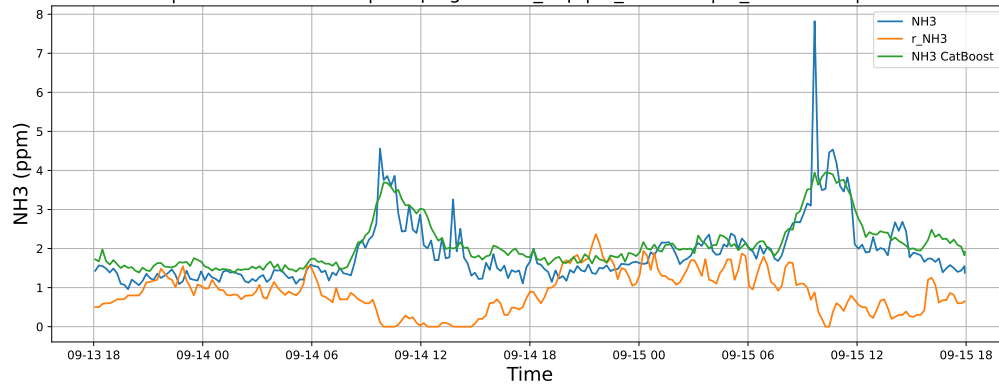
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

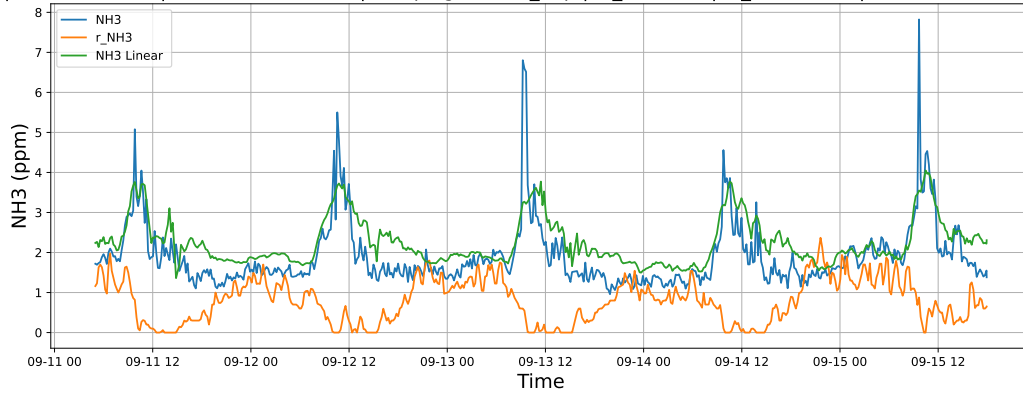


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

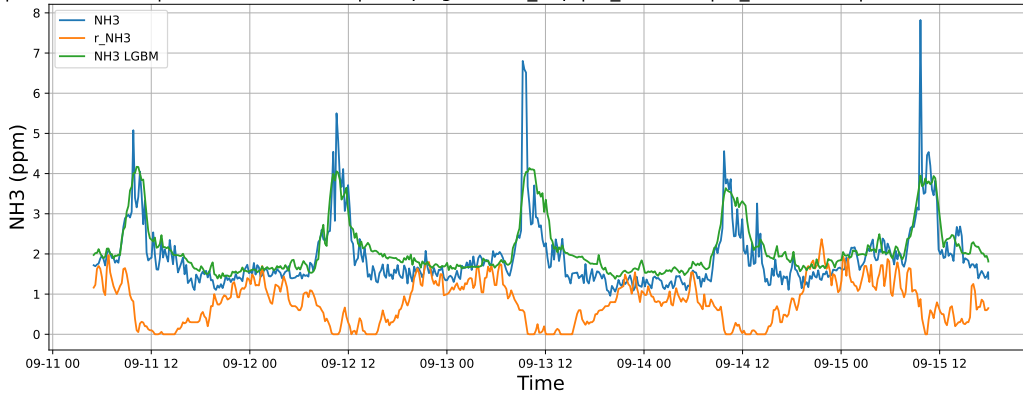


4.1.6.3 Time window : ALL

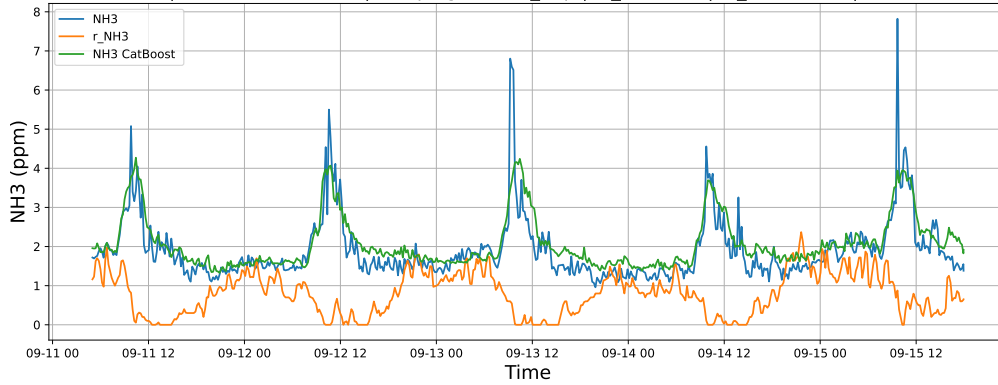
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



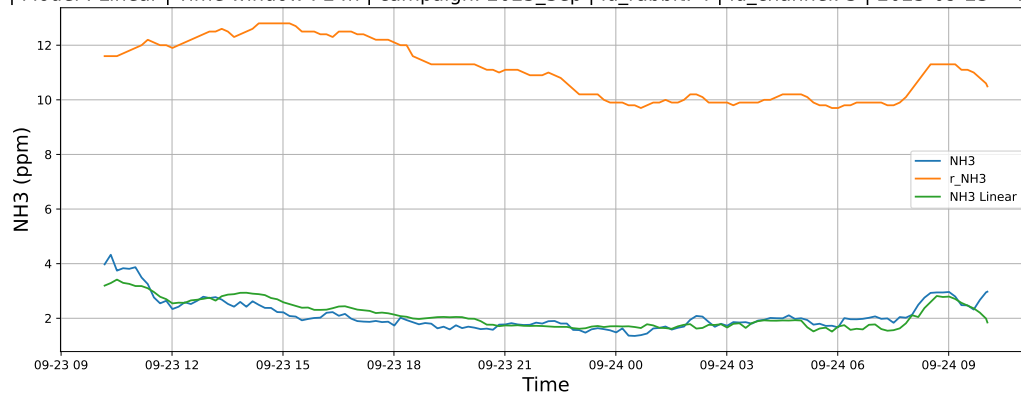
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



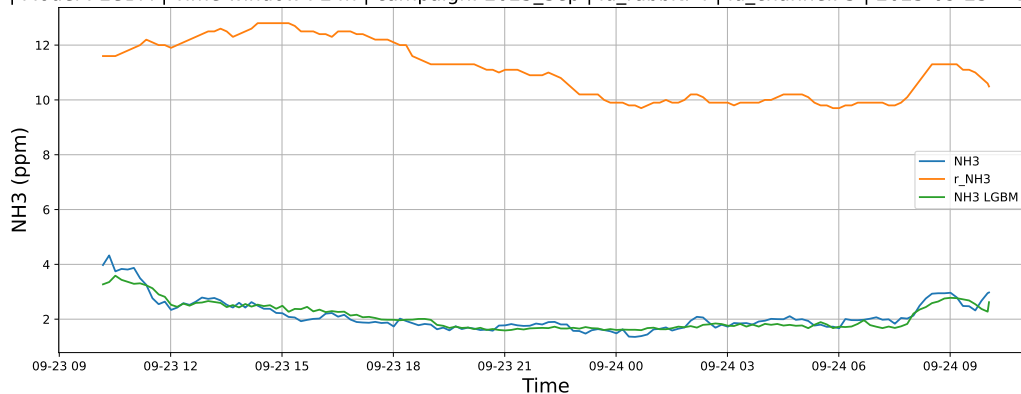
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

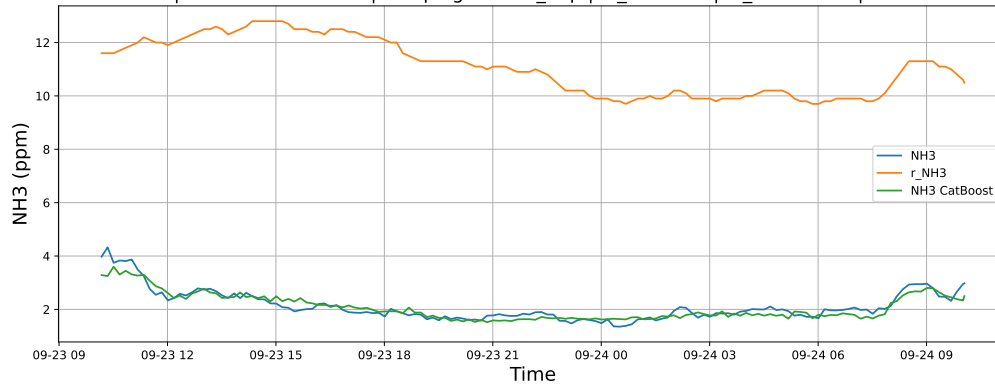
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

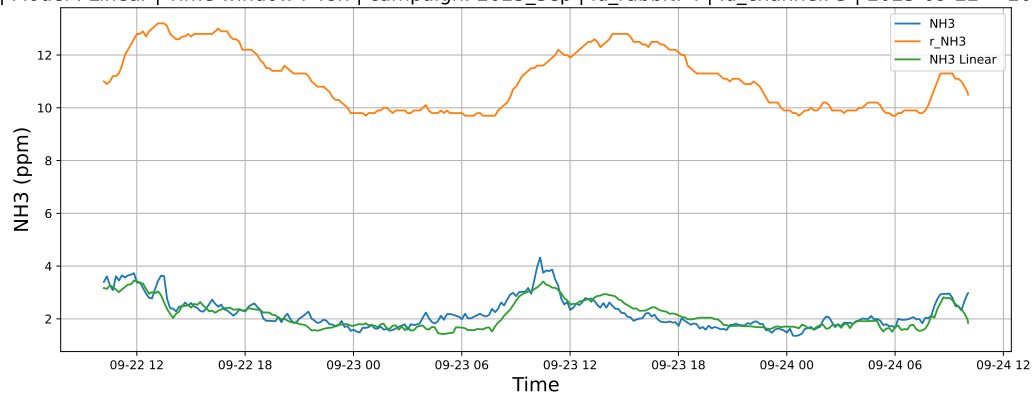


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

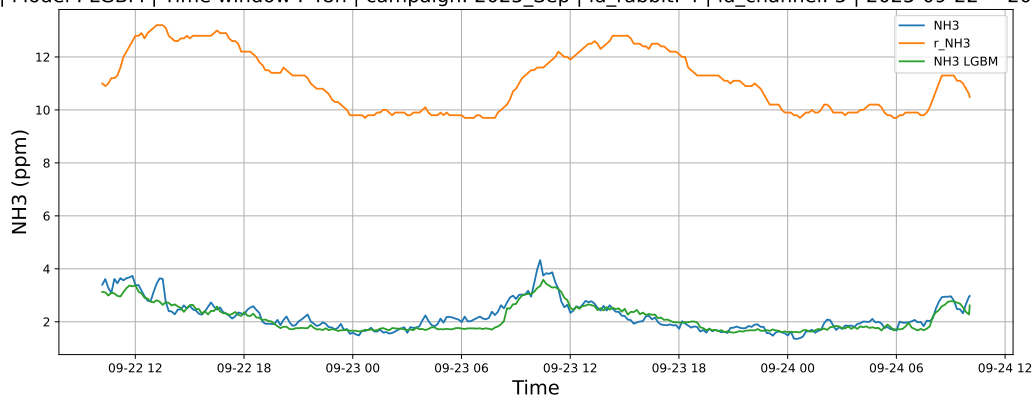


4.1.7.2 Time window : 48

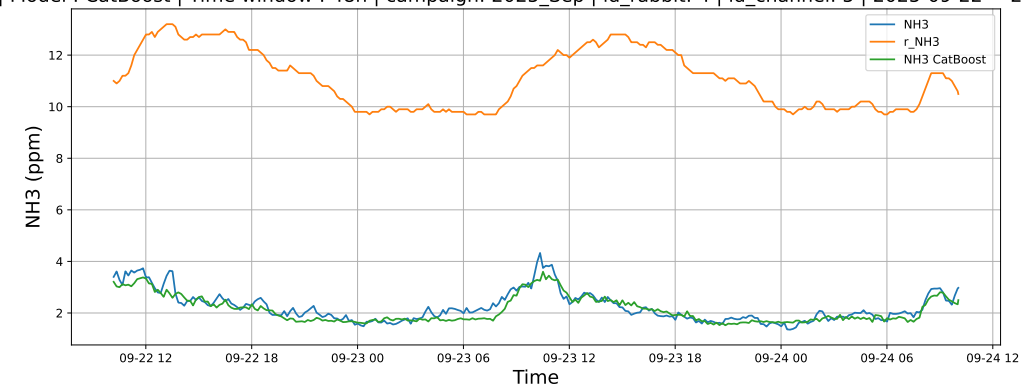
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

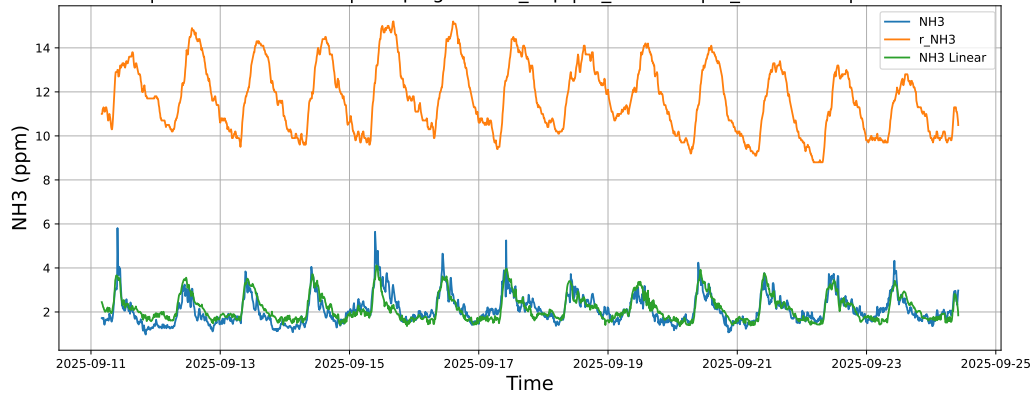


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

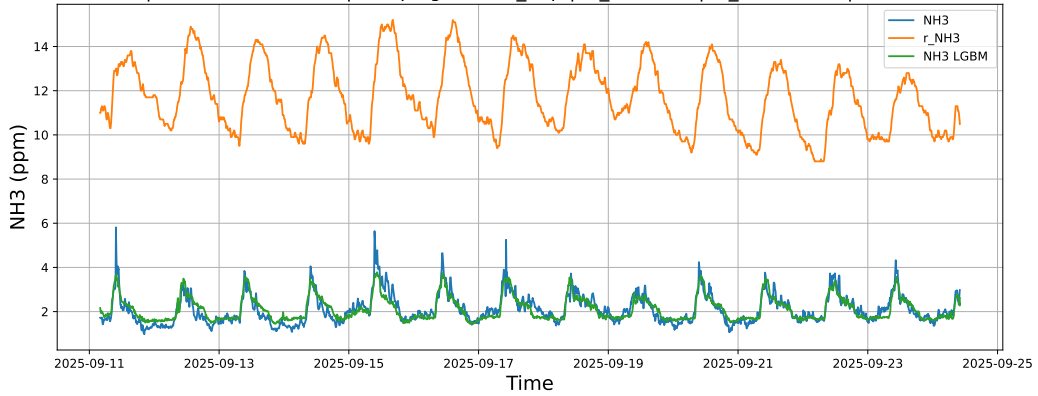


4.1.7.3 Time window : ALL

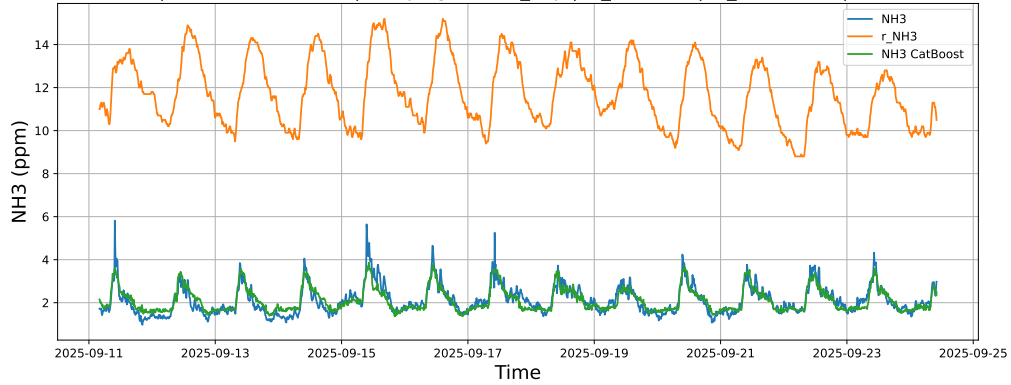
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



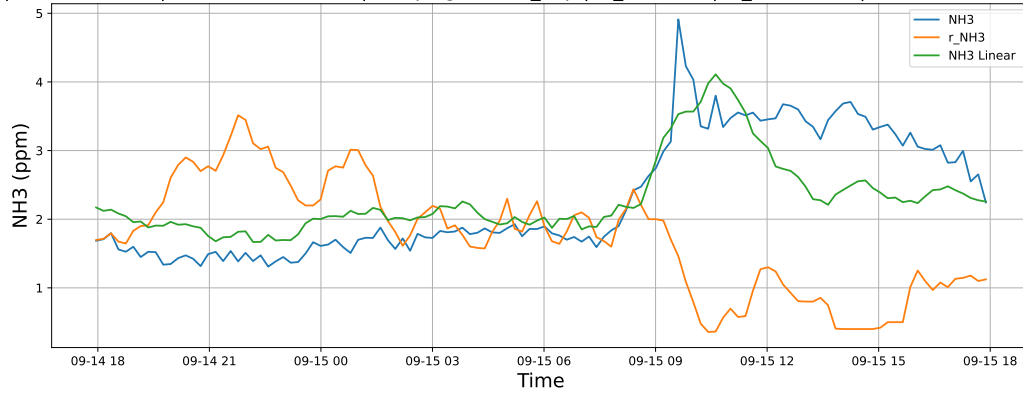
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



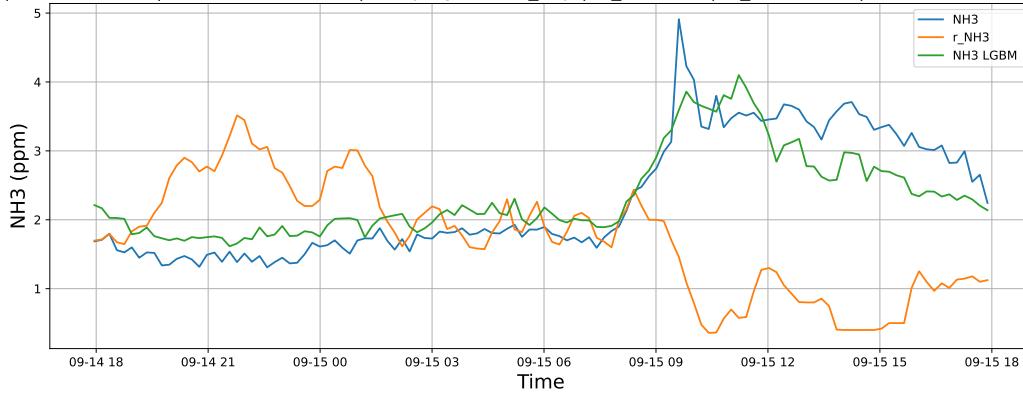
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

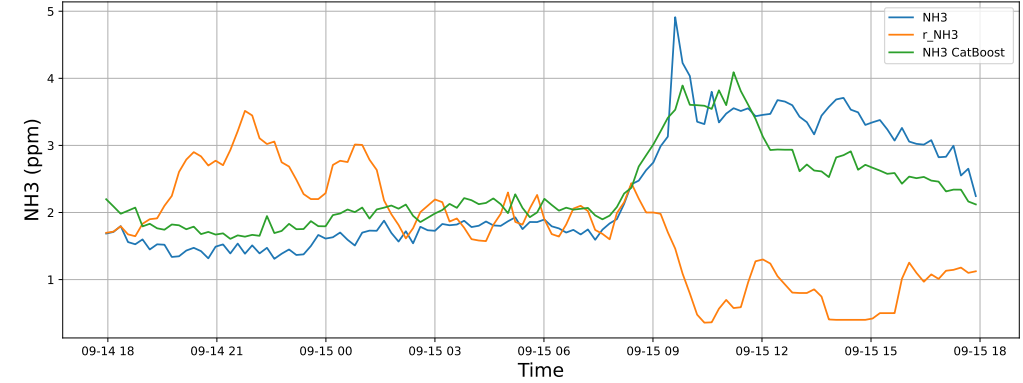
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

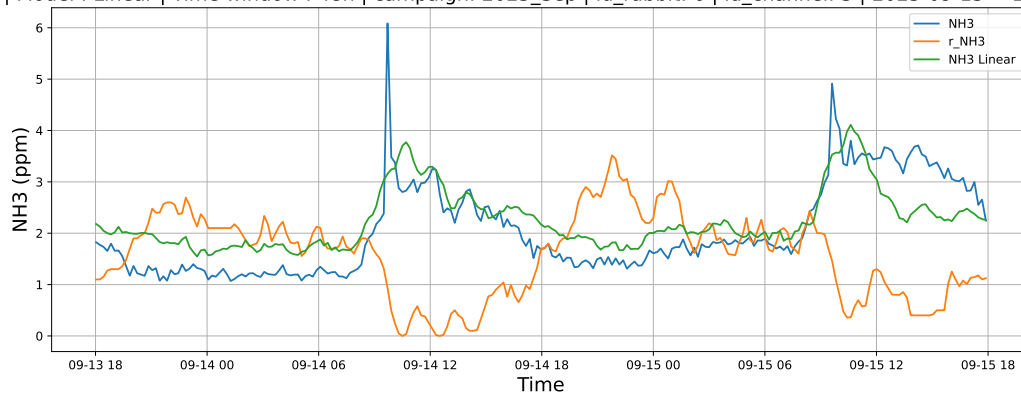


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

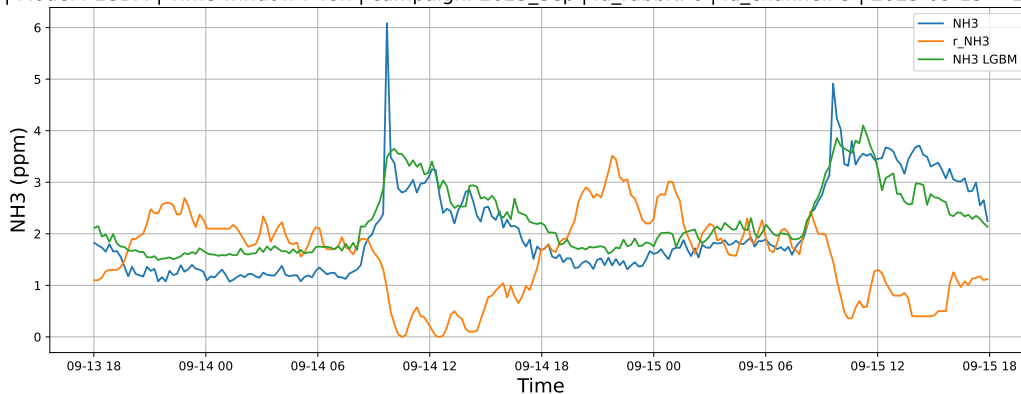


4.1.8.2 Time window : 48

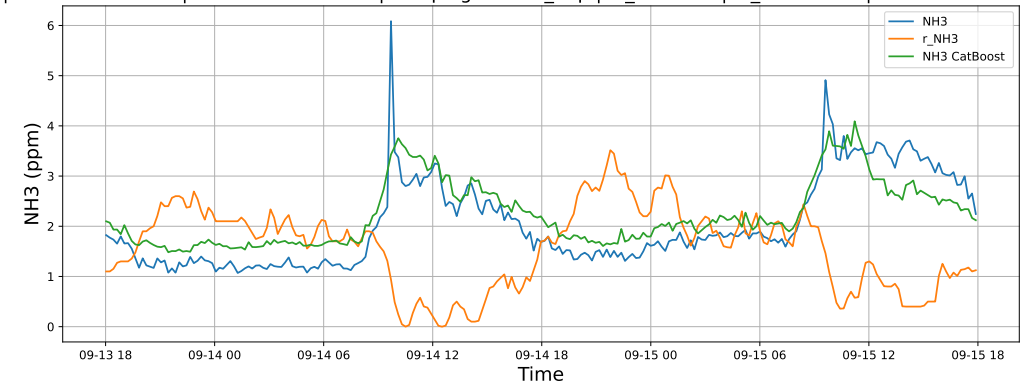
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

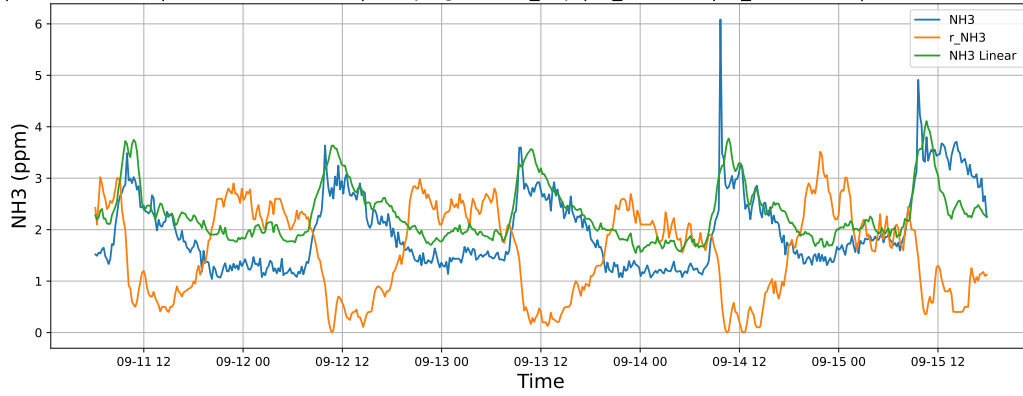


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

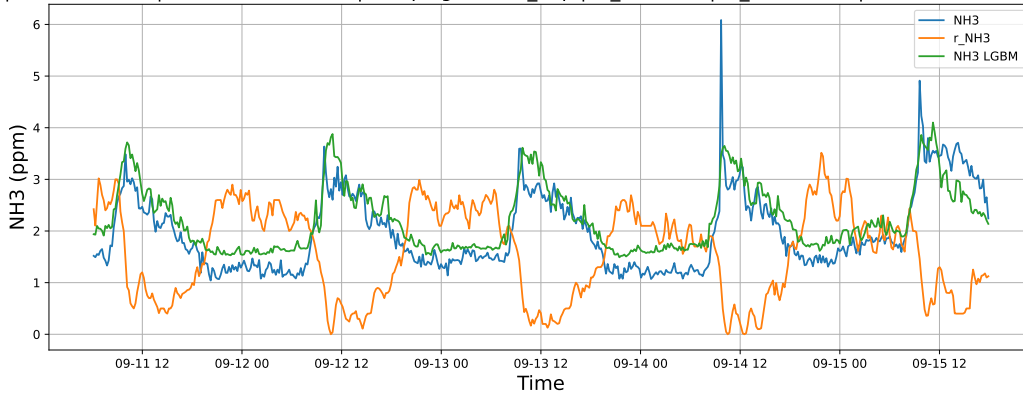


4.1.8.3 Time window : ALL

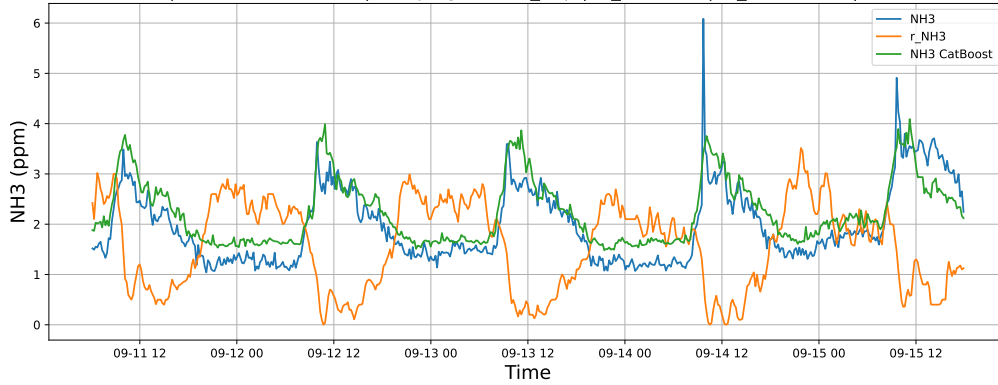
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

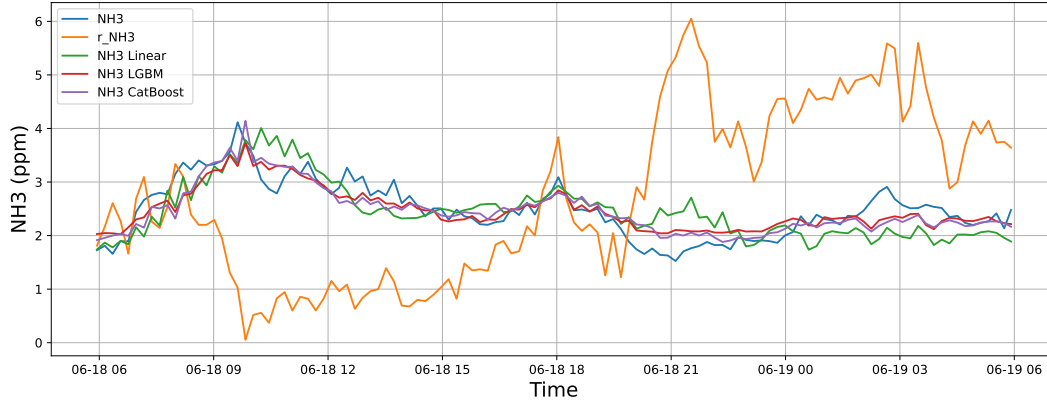


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

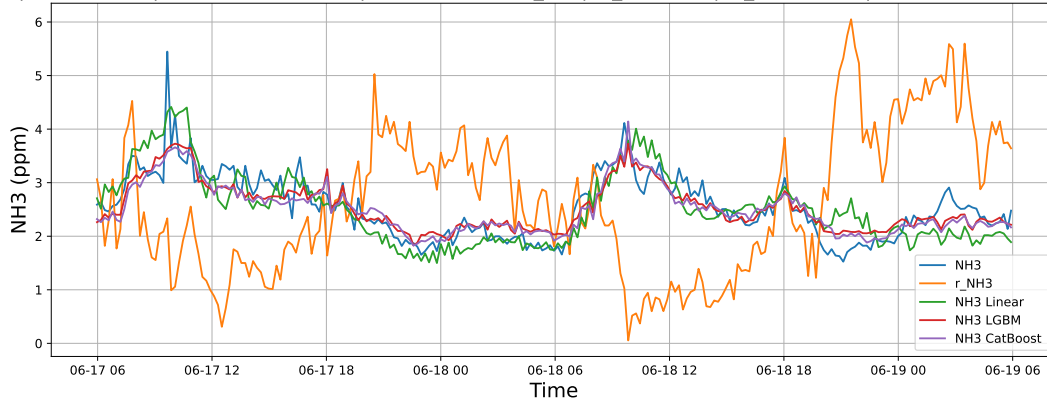
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



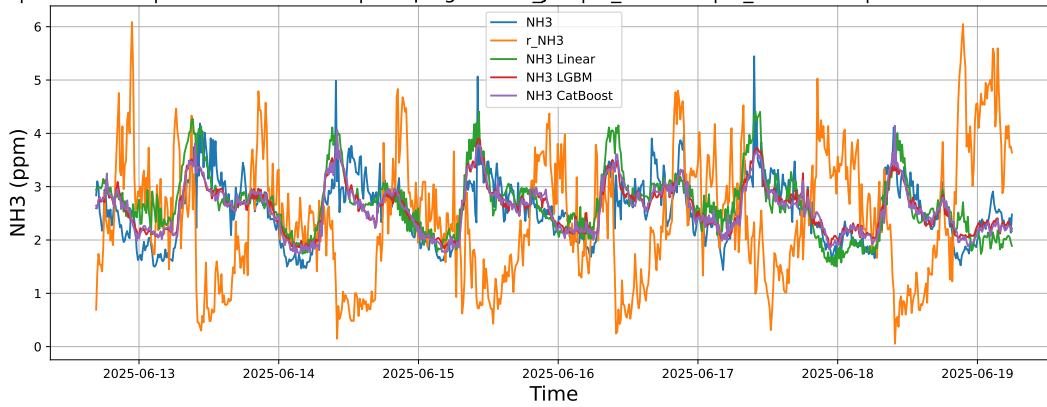
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

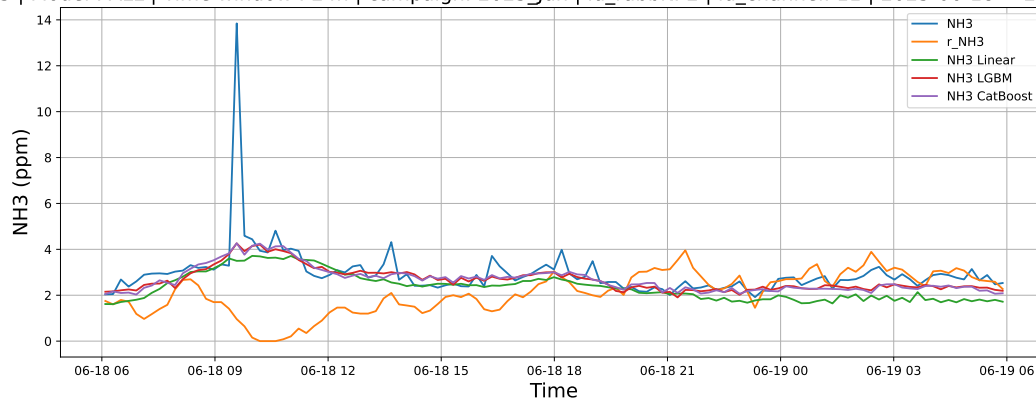
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

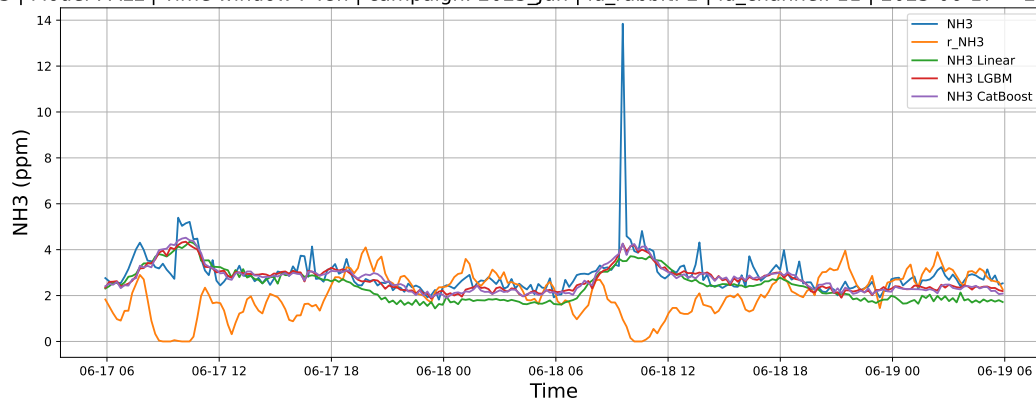
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



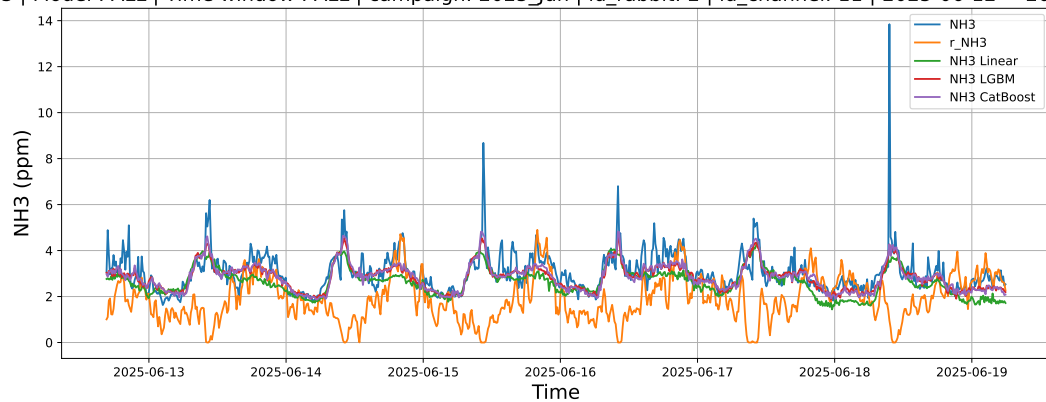
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

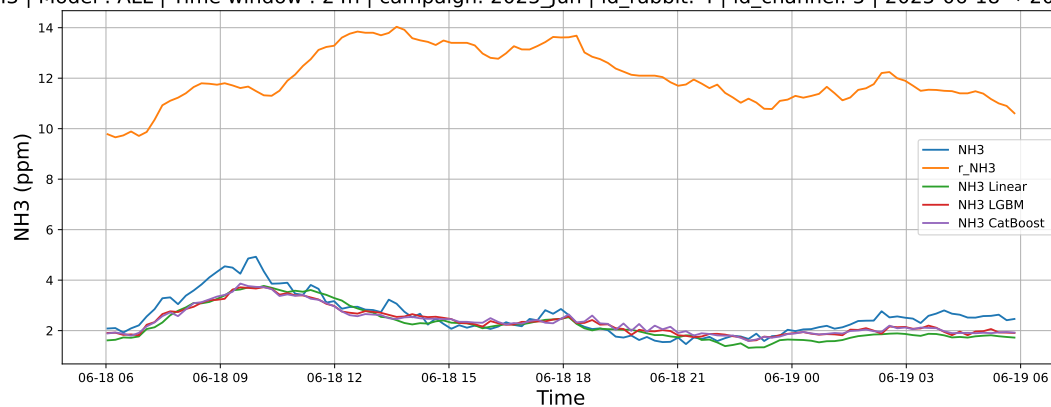
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

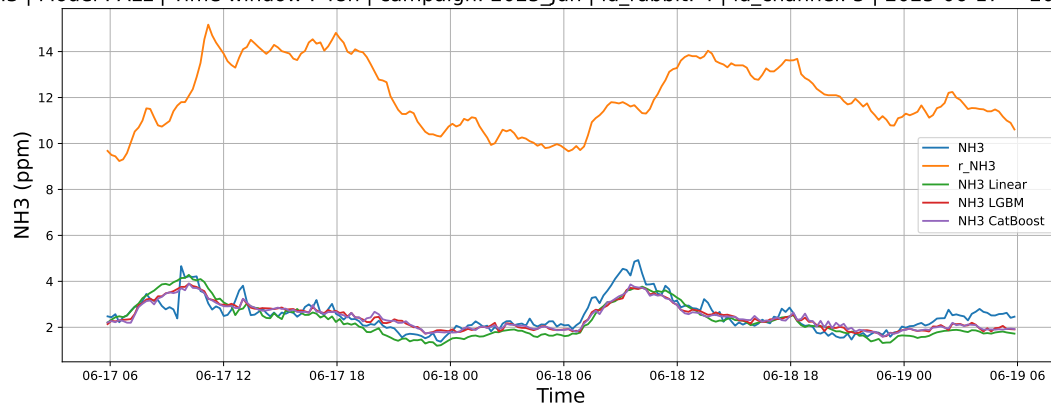
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



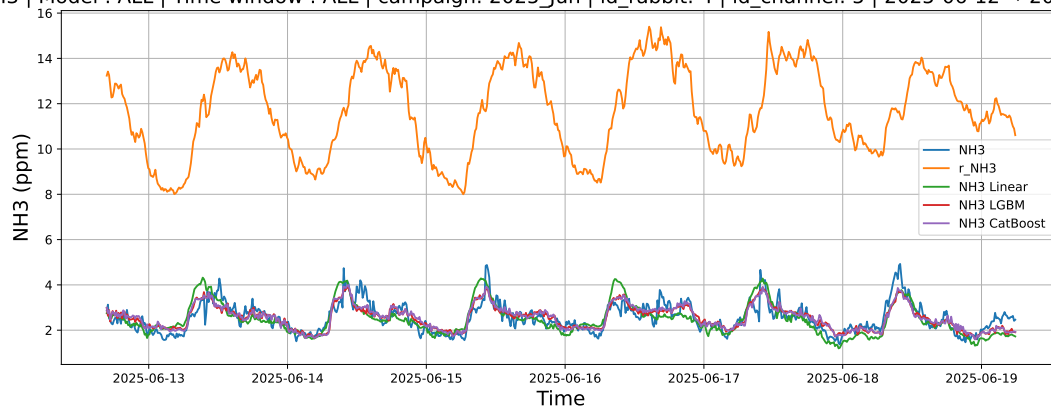
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

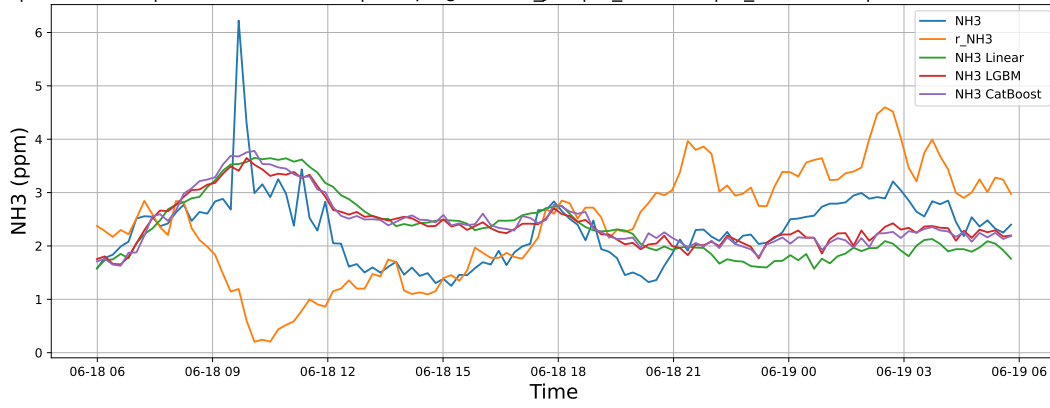
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

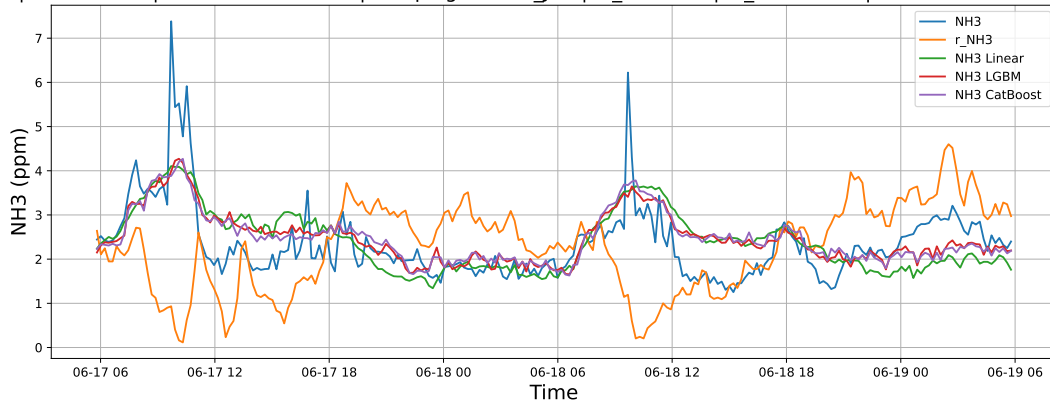
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



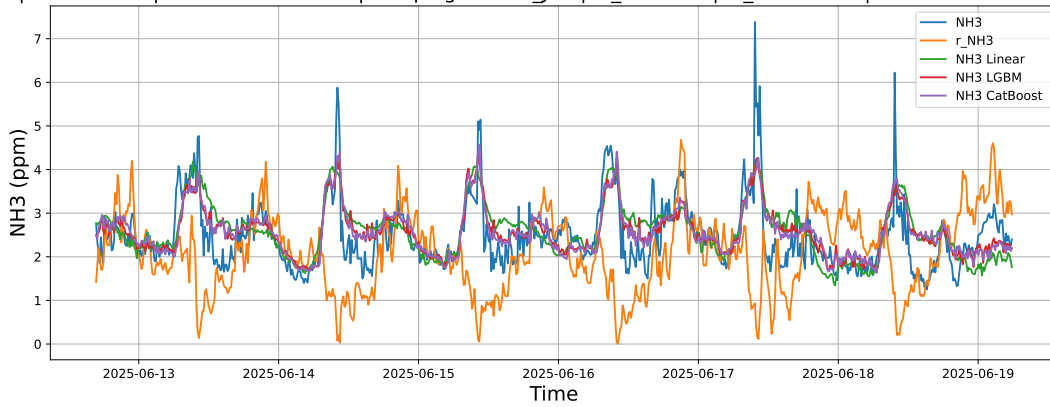
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

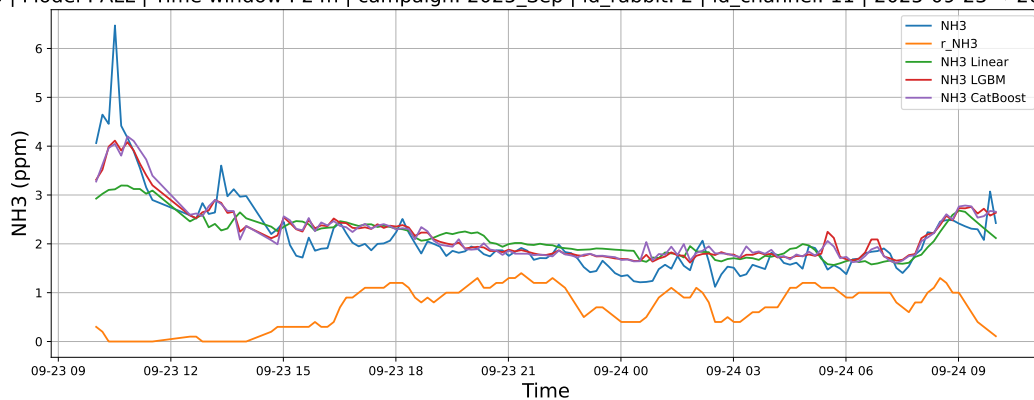
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

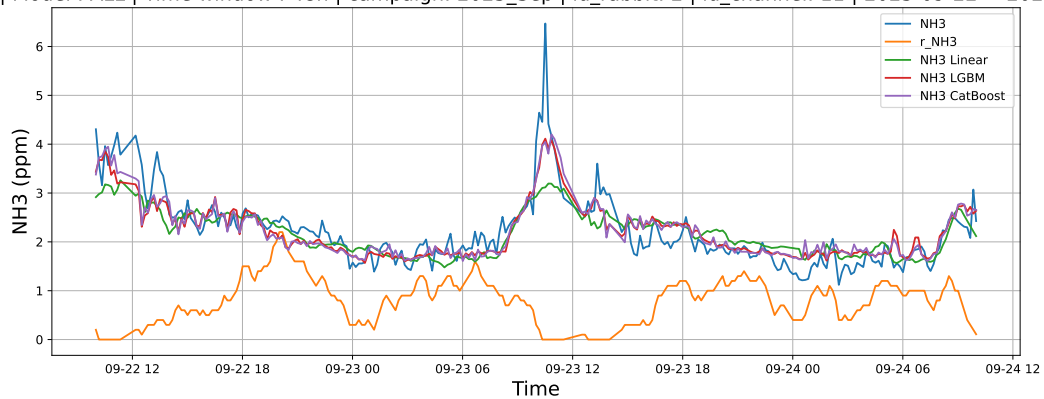
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



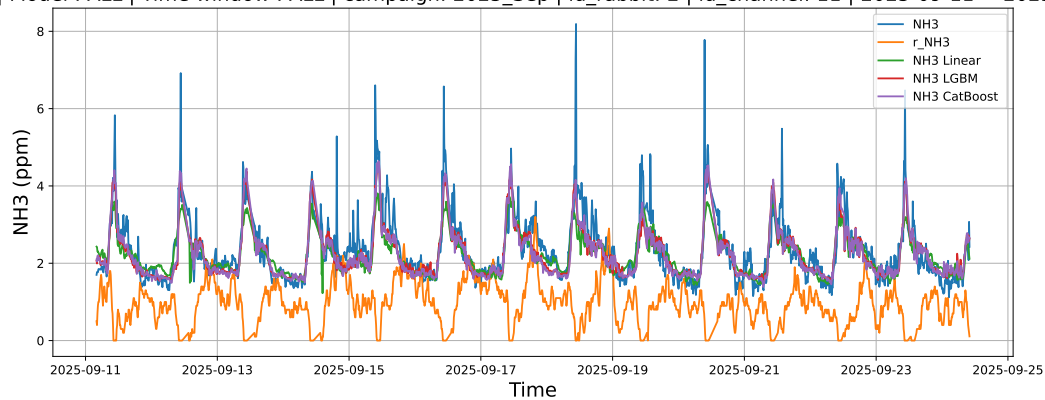
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

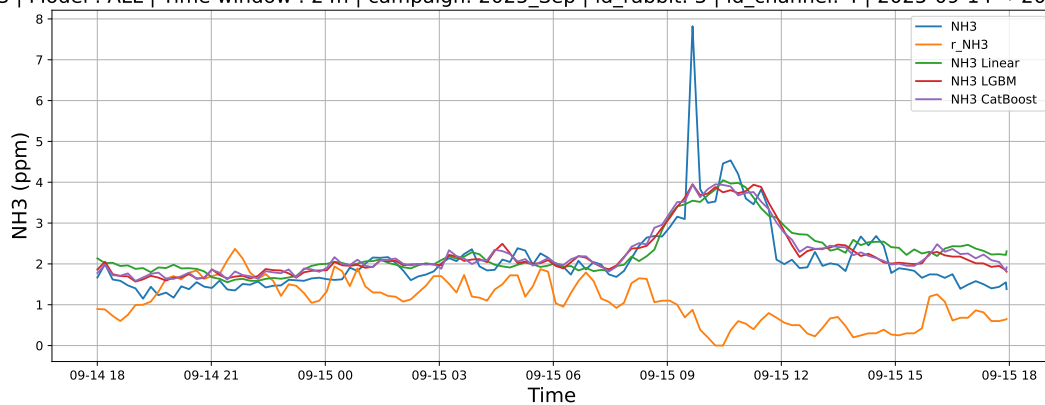
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

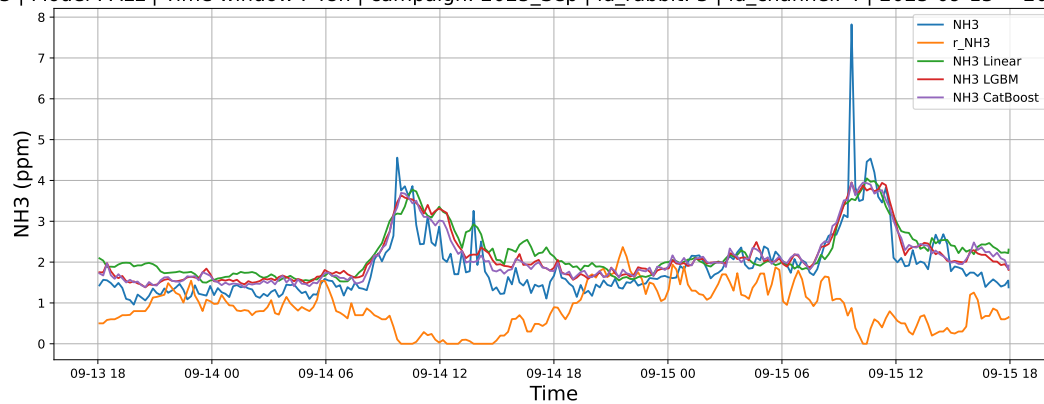
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



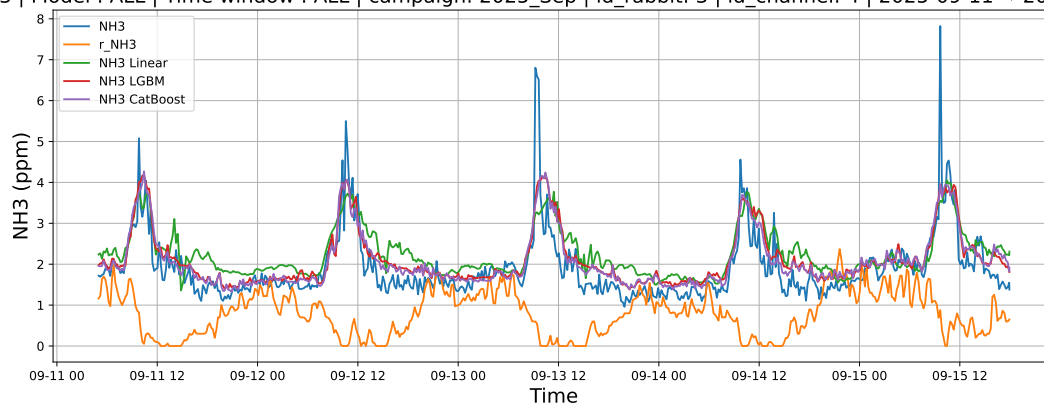
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

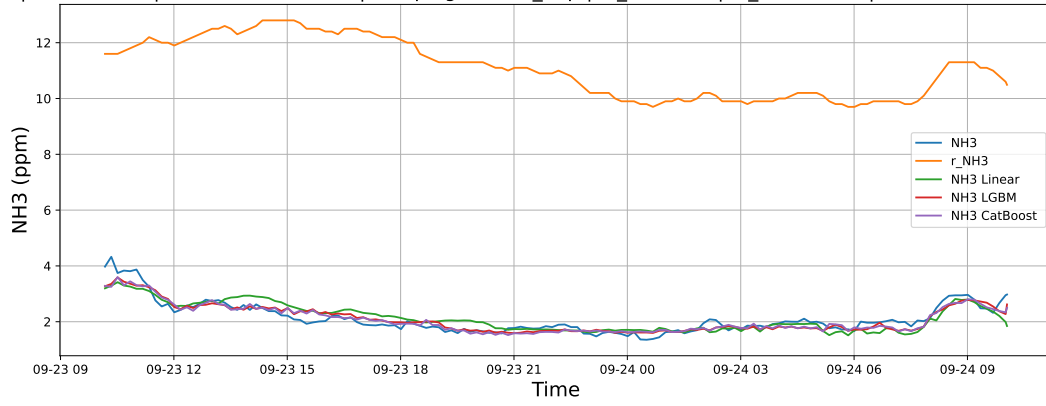
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

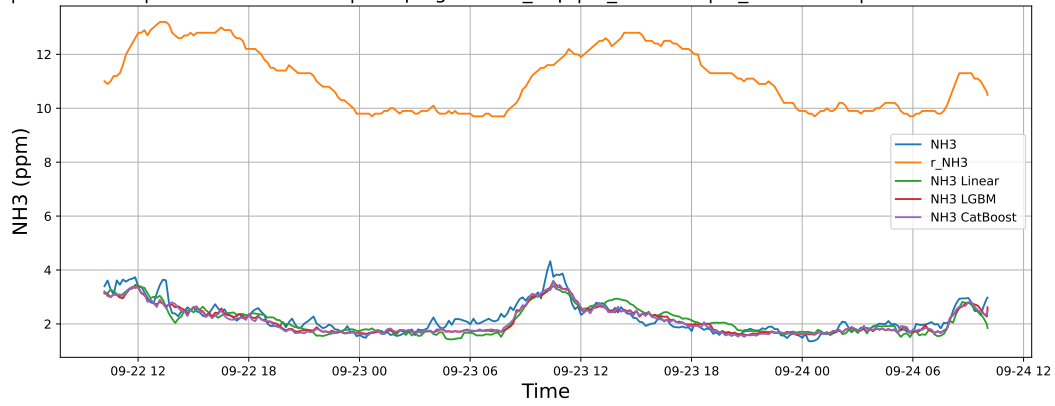
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



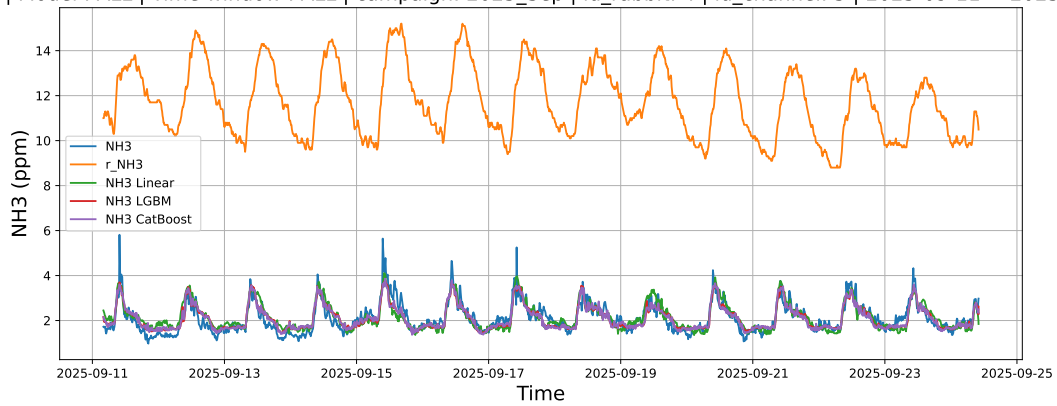
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

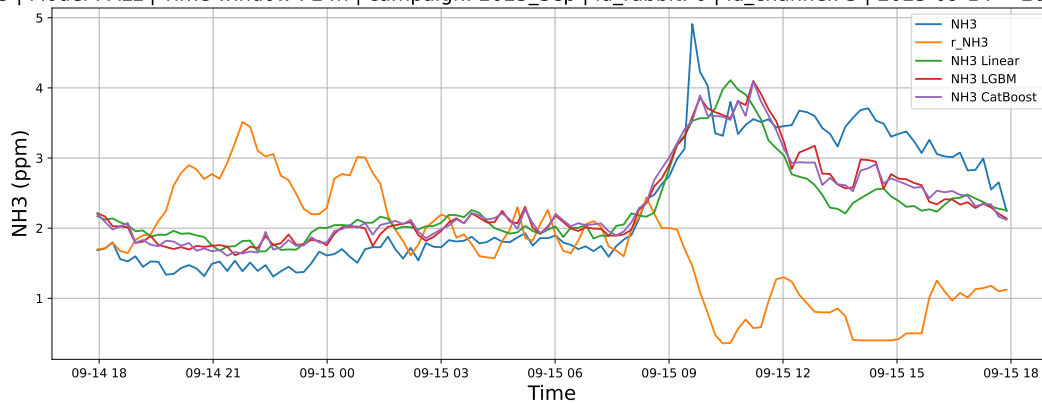
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

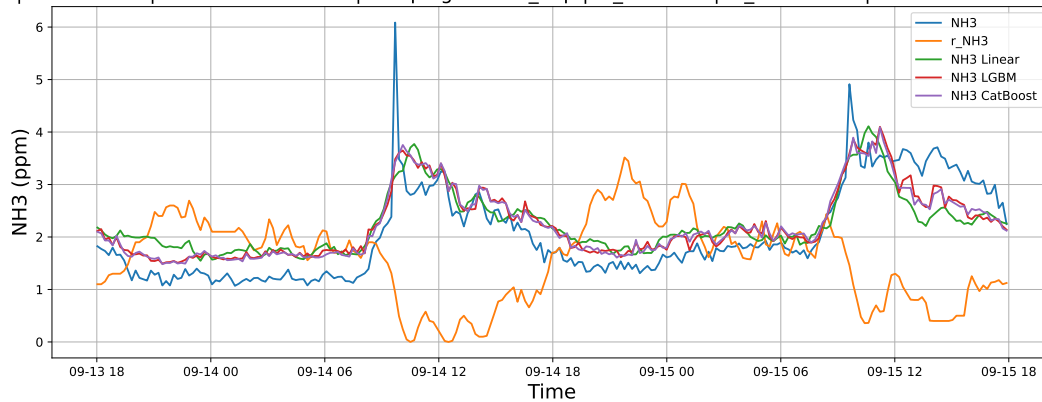
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

